

Entry Delays and Fighting Brands: Evidence from Generics and Authorized Generics*

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January 7, 2025

Abstract

Generic entry – a key driver of lower pharmaceutical prices and improved access – is slowed by various regulatory delays and deterred by Authorized Generic (AG) launch by the branded drug manufacturer. AGs are chemically identical to the brand drug, does not carry the brand label, priced lower than the branded drug, and can be released at any time. Using data on the US, we develop and estimate a structural model of generic entry and AG release timing to study counterfactuals. Promoting faster generic entry leads to fewer generic entrants yet lower prices in the early periods of the market. Banning AG leads to more generic entrants but mixed effects on prices – prices are higher in early periods due to missing AG competition, but may be lower in later periods if multiple new generics enter.

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1 Introduction

In pharmaceutical markets generic competition lowers drug prices and spending, thereby improving access and health outcomes for consumers. Understanding the economic mechanisms governing generic entry is important for crafting policies. However, generics face multiple hurdles when considering entry. First, generics require approval from Food and Drug Administration (FDA) to launch, and the time to get approval is uncertain and lengthy due to various regulatory delays. Second, they face strategic responses from the branded drug manufacturer (incumbent) that reduce generic profits. One controversial strategy of the incumbent is to release an Authorized Generic (AG). AGs are chemically identical to the branded drug but do not carry the brand label, are priced lower than the brand, and (unlike generics) can be released at any time. Such a strategy is known as “fighting brand strategy”.¹ This (potential) additional competition from a similar product worsens the prospects of generics considering entry. Such regulatory costs and uncertainty for entrants as well as incumbent’s fighting-brand response are common in many other industries.

In this paper we seek to understand how regulatory delays and AG release affects generic entry decisions and market outcomes. First, we provide descriptive evidence about generic entry and AG release. Second, we examine the impact of reduced regulatory delays for generics on market outcomes. Third, we illustrate the impact of AGs on short-term and long-term market outcomes, particularly generic entry deterrence. The latter can inform policy discussions about encouraging versus discouraging fighting-brand strategies and other product-line extensions. Controversies about AG release led to a report by the FTC (FTC (2011)) and are summarized in Feldman (2021) and various media coverage. While AGs can provide additional price competition, the key uncertainty in all these analyses is the impact of AGs on generic entry deterrence, which our paper aims to address.

To simulate such policies, we develop a stylized structural model of generics making entry decisions followed by branded drug manufacturer deciding when to release the AG. Regulatory delays manifest as generics receiving FDA approval stochastically over time,

¹In a fighting brand strategy, the incumbent releases a low brand-value version of its existing product, called a fighting brand. The high brand-value product charges high price and the fighting brand charges low price. This segments the market, with fighting brand competing with new entrants and original product serving the higher end of the market. An example of fighting brands (from Bourreau et al. (2021)) is Intel with the Pentium series (brand) and Celeron (fighter brand) to compete with AMD.

rather than being able to launch immediately. The model is estimated, then used to conduct policy simulations. Reducing regulatory delays leads to fewer generic entrants, yet prices are lower in the early periods of the market as all generic entrants launch almost immediately. However, the decline in generic entrants may lead to weakly higher prices in later periods of the market. Next we illustrate the impact of AG by comparing AG release with a hypothetical AG ban. The ban on AG leads to weakly more generic entrants, and sometimes multiple additional generics. The resulting impact on prices is mixed – prices are higher in early periods due to missing AG competition, but may be lower in later periods if multiple new generics now choose to enter.

An outline of our structural model is as follows. The demand side is a model of consumers choosing between brand, AG, and generics. The supply side of the industry is modeled in two stages. First, generic manufacturers decide whether to apply for entry into a market. Second, a dynamic game begins where every period, generics who applied to enter receive approval stochastically, the branded drug manufacturer decides whether to release an AG, and market participants compete in prices. This two-stage model setup reflects the institutional reality that unlike AGs, generic entry faces uncertain regulatory delays (and thus generics need to apply for entry in advance). We solve the two-stage supply model by backward induction, and as a result allow for AG and generics to form expectations about each others’ entry and pricing decisions when making a choice.

We use data on national sales and revenue for prescription drugs in the US between 2004-2016 (IQVIA National Sales Perspective) to estimate the structural parameters. We define “loss-of-exclusivity” to be the first quarter when any generic enters a molecule-formulation. Our empirical measure of regulatory delays is estimates of average generic entry rates for every quarter since loss-of-exclusivity; these are used by generics to form expectations about entry dates. Prices in different industry states are predicted using a reduced-form regression to avoid making pricing conduct assumptions. We follow previous papers in defining a market as a molecule-formulation ([Conti and Berndt \(2020\)](#), [Wang et al. \(2022\)](#), [Yurukoglu et al. \(2017\)](#)).

For our counterfactuals, we solve the model for each market. An important element of market dynamics is that the early quarters of a market are most profitable for generics and AGs – there are few generics present as the rest wait to get FDA approval, resulting in lower price competition and higher market shares.

Our first set of counterfactuals illustrate how both generics and AGs can deter each other's entry, but in different ways. Generics can deter AG because they move first, i.e. they apply for entry and incur the entry cost first. However, AGs can also deter generics because they can launch their product earlier than generics – the early launch significantly reduces generics' expected profits by providing price competition in the early profitable periods of the market. To showcase these two forces, we first show that no AG is released when the generics can launch immediately and the AG has no cost or demand advantage; we then show that AG is able to deter more generics when it can launch earlier rather than being forced to delay.

Our second set of counterfactual simulations study the impact of reducing regulatory delays to promote faster generic entry. This has been a priority of the FDA since the 2010s. We find that reduced regulatory delays leads to weakly fewer generic entrants. This is because while reduced delays means each generic operates for more periods, its rivals are also able to launch earlier in the market. Each generic therefore faces intense price competition as soon as it enters, and this more than offsets the advantage from operating for longer periods. Lifetime generic profits fall, and thus fewer generics choose to enter.² Despite fewer generic entrants, prices are lower in early periods of the market. This is because while there are fewer generics, they all launch almost immediately, again leading to intense price competition from the beginning. However, fewer generics may result in higher prices in later periods.

Our third set of counterfactuals contrasts the presence versus absence of AG on short-term and long-term market outcomes. This can inform policymakers about whether AGs should be encouraged or banned. We show that a ban on AG generally leads to more generic entrants, and in fact may lead to multiple additional generics. The AG ban can incentivize multiple new generic entrants – its early-launch advantage means that it is present earlier than most generics and often during the most profitable periods of the market. As a result, its presence has a more negative impact on a generic's lifetime expected profits than other independent generics. The effect of AG ban on prices is mixed. When an AG is released,

²Another way of seeing this is that increasing regulatory delays creates an incentive for an additional generic entrant to take a gamble and apply for entry – while it makes the same per-period profit in the long run with or without regulatory delays, there is now a chance that its rivals will be slow to obtain approval while it gets early approval. This leads to higher profits for the generic in the short run when compared to the case with no regulatory delays.

it is usually in the early periods of the market just after loss-of-exclusivity. Therefore, with the AG ban, generics entering the market early do not face an additional competitor through the AG; thus, prices are higher in the early periods of the market. However, if the AG were deterring multiple generics from entering, then prices in the later periods of the market would be lower with the AG ban as multiple new generics choose to enter.

Contributions and related literature: This paper contributes to the following strands of literature. First, we contribute to a small literature on Authorized Generics. We are the first to study the AG release decision and its impact on generic firms by using a structural model of entry. [Appelt \(2015\)](#) uses a recursive bivariate probit regression to show that AG entry does not impact generic entry in Germany. A report from the Federal Trade Commission ([FTC \(2011\)](#)) leveraged many information sources to explain the decision to release the AG and generic manufacturers' reactions to it. [FTC \(2011\)](#) mostly focuses on how AG release may disincentivize Paragraph IV lawsuits by generics and does so by compiling industry data and company documents; this paper instead focuses more generally on AG release, and considers Paragraph IV certifications and settlements as exogenous to our model. Both [FTC \(2011\)](#) and this paper find that the majority of AGs are released in the absence of a Paragraph IV lawsuit. Our descriptive findings broadly mirror those of [Fowler et al. \(2023\)](#), who use industry data to describe broad trends about Authorized Generics over 2010-2019; see Appendix [G](#) for a comparison.

Second, we construct a simple structural model of generic industry and pharmaceutical product entry that captures generic industry dynamics, yet is quick to solve and can be easily extended in various directions. While our structural model is strictly about AGs, we believe it can be extended to study other strategic reactions against generics by branded drug manufacturers (product line extensions, REMS requirements, orphan drug classification, etc.) and other regulatory delays (such as those faced by biosimilars). In particular, the model can be used to tractably study generic entry deterrence. In doing so, we add to the literature on generic drug entry and deterrence. In contrast to other papers, we develop a parsimonious model that captures generic industry dynamics yet is easy to solve and extend. Other notable papers studying generic entry via structural models are [Ching \(2010\)](#), [Gallant et al. \(2017\)](#), [Scott Morton \(1999\)](#), [Reiffen and Ward \(2005\)](#), [Starc and Wollmann \(2023\)](#), and [Wang et al. \(2022\)](#); see [Ching et al. \(2019\)](#) for an overview. More generally, our paper relates to a large literature on generic competition and pharmaceutical

pricing; key references for our paper include [Arcidiacono et al. \(2013\)](#), [Berndt et al. \(2007\)](#), [Bokhari and Fournier \(2013\)](#), [Chen et al. \(2022\)](#), [Dubois et al. \(2022\)](#), [Ellison and Ellison \(2011\)](#), [Frank and Salkever \(1992\)](#), [Frank and Salkever \(1997\)](#), [Ganapati and McKibbin \(2023\)](#), [Hermosilla and Ching \(2023\)](#), [Lakdawalla \(2018\)](#), [Olson and Wendling \(2018\)](#), [Reiffen and Ward \(2007\)](#), and [Tenn and Wendling \(2014\)](#); see [Morton and Kyle \(2011\)](#) for an overview. [Berndt et al. \(2017\)](#), [Berndt et al. \(2018\)](#), and [Conti and Berndt \(2020\)](#) provide important descriptive evidence about the aggregate dynamics and inner workings of the generic industry which we use to motivate our model.

Third, we contribute to a somewhat limited empirical literature on entry deterrence and preemption. Unlike most papers in this area, we tackle empirical issues regarding entry deterrence differently. The fundamental empirical problem is that when a potential entrant does not enter, the researcher cannot tell if this was due to deterrence or due to absence of entry threat to begin with (see [Cookson \(2017\)](#) and [Gil et al. \(2024\)](#) for a detailed discussion and creative workarounds). In our paper, we tackle this by building a model of entry decisions, and using it to predict entry decisions in different markets. The sequential nature of moves in this institutional setting (i.e. generics make entry decision first followed by AG) leads to a clean solution of the entry deterrence problem. Empirical papers on this topic include [Cookson \(2017\)](#), [Fang and Yang \(2024\)](#), [Gil et al. \(2024\)](#), [Goolsbee and Syverson \(2008\)](#), [Igami and Yang \(2016\)](#), and [Schmidt-Dengler \(2024\)](#). There is an extensive theoretical literature on this topic, which can be found in the references above.

Fourth, we contribute to a very sparse Empirical IO literature on fighting brands. We are one of the very few papers to build and estimate a model of an incumbent releasing a fighting brand. Fighting brands are widely observed in many industries besides pharmaceuticals, so our model and results add to our understanding of such strategies, particularly entry deterrence motives. [Bourreau et al. \(2021\)](#) study how fighting-brand launches marked a breakdown of collusion in French mobile telecommunications. There is a theoretical literature on fighting brands, notably [Anderson and Dana \(2009\)](#), [Deneckere and McAfee \(1996\)](#), and [Johnson and Myatt \(2003\)](#).

Finally, our model combines elements from the Empirical Industrial Organization literature on static entry games and dynamic games. For static entry, key references include [Berry \(1992\)](#), [Berry et al. \(2016\)](#), and [Seim \(2006\)](#). For dynamics, some key references for our setting are [Gowrisankaran and Rysman \(2012\)](#), [Igami \(2017\)](#), [Igami and Uetake](#)

(2020), and [Rust \(1987\)](#); see [Aguirregabiria et al. \(2021\)](#) for a detailed overview of the literature.

2 Institutional Background

To enter a market after the branded drug manufacturer loses exclusivity, generics need to apply for FDA approval. However, gaining FDA approval is lengthy, costly, and takes an uncertain amount of time. The brand drug manufacturer can respond to generic entry by releasing an Authorized Generic (AG). The AG is chemically identical to the brand drug but without the brand label attached. Unlike generics, the AG can be launched at any time without having to wait for FDA approval. The general patterns after loss of exclusivity are: branded drug prices are high, generics and AG charges low price which further declines with more competition, and the branded drug loses most of its market share to generics and AGs over time.

2.1 Generic Drugs and Market Entry

A pharmaceutical manufacturer invests in R&D to discover a drug, then patents the molecule and conducts tests for efficacy and safety. This drug is commonly referred to as the “branded drug” (see [Dubois et al. \(2015\)](#) and [Khmelnitskaya \(2023\)](#), and the references therein). Throughout we refer to the manufacturer that pioneered the molecule as the “branded drug manufacturer” and the pioneer molecule as the “branded drug”. We use the term “nonbrand” to indicate both generics and AGs.

After loss-of-exclusivity (LOE) of the branded drug, generic drugs can enter into the molecule-formulation once they receive final approval from the FDA. The FDA’s approval process is costly for the generic manufacturer, lengthy, and has uncertain duration.

The generic manufacturer submits an Abbreviated New Drug Application (ANDA) to the FDA following a process laid out in the Hatch-Waxman Act. The mean approval time for an ANDA is between 32-40 months for our data period, so generics typically submit their ANDAs well in advance of LOE. Approval times of generics vary due to the amount of work involved in verifying the evidence submitted in support of the entrant’s ANDA, as well as the FDA’s workload. The FDA has experienced a significant backlog of pending

ANDA applications. Policies such as the GDUFA – and its reauthorization every 5 years – have given the FDA more resources to reduce the time for generic drug approvals (Berndt et al. (2018), Wang et al. (2022)).

To gain FDA approval, generic manufacturers need to prove to the FDA that their products are “bioequivalent” to the branded drug and can be produced at scale. Bioequivalence is defined to mean “product has the same active ingredient, dosage form, strength, route of administration and conditions of use” as the reference branded drug.³ Generic manufacturers have to give further evidence that they can manufacture the drug accurately, at scale for commercial distribution, and according to Good Manufacturing Practices.⁴ This might further involve a back and forth between the FDA and generic applicant until the FDA is satisfied with the production line. An important aspect of proving bioequivalence is that the generic firm needs to reverse-engineer the branded drug. Since some aspects of the manufacturing process for the branded drug may be undocumented or kept as trade secrets by the branded manufacturer, generic firms must experiment with production to meet regulatory requirements. The cost incurred to receive FDA approval range from \$3 million to \$15 million.⁵

In this paper, we refer to applying for FDA approval as “applying for entry”, and refer to generics launching their product after FDA approval as “entry”.

Generics face a variety of regulatory delays that slow down entry into a molecule-formulation. Some examples are as follows. First, the most important is the time taken by the FDA to assess ANDA applications and give approval, which often involves a back-and-forth between the FDA and the manufacturer regarding the production process. Over our dataset’s time periods, the FDA suffered from large backlogs of ANDA applications waiting to be approved, although this has improved. Second, the Hatch-Waxman Act set up a 180-day first-filer-exclusivity provision where the first generic to file for Paragraph

³Bioequivalence ensures that generic drugs are not lower quality compared to their equivalent branded drugs - by definition, their bioequivalence means that they can be substituted with the branded product and have the same effect. Thus, generic drugs compete with their branded equivalents in the US on the basis of their availability and price. Source: <https://www.fda.gov/media/71401/download>

⁴See information from the FDA on Good Manufacturing Practices: www.fda.gov/drugs/pharmaceutical-quality-resources/current-good-manufacturing-practice-cgmp-regulations.

⁵A range of generic entry cost estimates can be found in Berndt et al. (2017), Gottlieb (2016), Scott Morton (1999), and Starc and Wollmann (2023), among others. The range we state is in the window of most reported estimates.

IV certification⁶ enjoys a 180-day period after entry when other generics cannot enter; thus, a rival becoming the first-filer stops other generics from entering the market for 180 days. Third, final approval by the FDA requires inspections to meet CGMP (Current Good Manufacturing Regulations) at scale, and they can fail. Furthermore, for certain types of molecules and formulations, the generic manufacturer can face manufacturing problems when setting up production lines for FDA approval, delaying approval further. Fourth, unrelated inspections for the first-filer can fail, which then cascades to interrupting final approvals for all subsequent filers. Finally, there may also be information issues due to legal or technical reasons such that the generic manufacturer cannot correctly anticipate the date of LOE, leading to delays in filing and hence approval.

2.2 Authorized Generics

In response to generic entry, the branded drug manufacturer can release an Authorized Generic (AG). The AG is chemically identical to the branded drug in terms of molecule structure and formulation, except that it does not have the brand label attached.

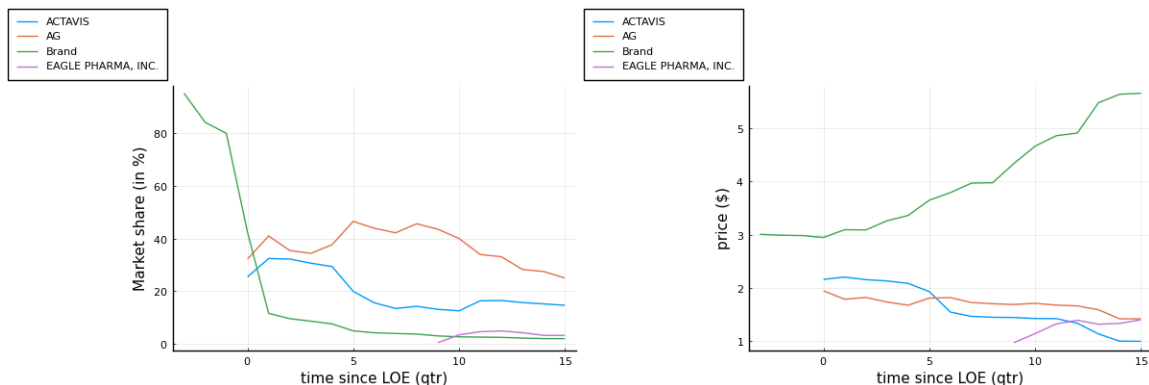
A key detail about product launch in this setting is that, unlike generics, AGs can be introduced at any time. This is because they are riding on the branded drug's approval from the FDA, and therefore do not require any additional approval.

Why would a branded drug manufacturer not release an AG every time it faces generic entry? There are several reasons. First, the AG could cannibalize the branded drug's sales. Second, the AG entry would lead to generics lowering their prices in response, further increasing the price gap between brand and generics products; this causes branded drug sales to fall even more. Third, it could be due to supply side reasons such as manufacturing delays, organizational frictions, and settlements. Fourth, it could be that many generics enter in the very first period after loss-of-exclusivity, meaning that the AG will make little profit due to intense price competition; this explains why AGs have a strong incentive to enter during the 180-day exclusivity period as detailed in [FTC \(2011\)](#). In our model we account for all these factors.⁷

⁶A Paragraph IV certification lawsuit by the generic manufacturer claims that the branded drug's patent is invalid, or will not be infringed upon by the generic manufacturer's product.

⁷This raises the question of why the brand manufacturer does not always match the price of generics, instead of releasing the AG. First, there is a price discrimination motive where consumers who like the branded drug are willing to pay a high price for it, so having the expensive brand and inexpensive AG in

Figure 1: Evolution of market shares (left) and prices (right) for Diclofenac-Misoprostol (oral formulation).



A brand manufacturer does not always directly release its own AG, instead it may be done by a generic subsidiary of the brand. It may also be done by firms which specialize in releasing AGs on behalf of other manufacturers such as Prasco. A generic manufacturer may also team up with a brand manufacturer to release its AG. In these cases, the non-brand firm takes over part of the responsibilities, and in return gets a share of the profits. See Appendix I for more details on this from [FTC \(2011\)](#).

2.3 Evolution of Market Shares and Prices

In terms of market prices and shares, the general pattern we see in the data is as follows. After loss-of-exclusivity by the branded drug, generics enter the market and undercut the brand's price, and the branded drug's price remains above that of generics. An AG is sometimes released by a branded drug manufacturer in response to generic entry. The AG and generics set prices close to each other. Over time, the branded drug's market share falls dramatically, while those of nonbrand drugs rises. Examples of this pattern can be seen in Figure 1 for a molecule-formulation.

the market allows the manufacturer to capture both types of consumers. Second, if the generic entrant has to leave the market (e.g. due to legal reasons, patenting issues, technical or manufacturing problems) the brand may not be able to raise its prices back up without facing backlash from buyers. Third, if the brand product's price is lowered, it is hard to justify high prices on line extensions. All the points mentioned above are corroborated in [FTC \(2011\)](#).

2.4 Controversies surrounding Authorized Generics

The practice of AG release has generated considerable controversy. It led to the FTC releasing a report in 2011 (FTC (2011)) studying the impact of AGs on market outcomes. The report concluded that while AGs reduce generic profits, they do not deter generic entry. Our paper revisits this question with better methodology. Much of the controversy about AGs is summarized in Feldman (2021). The issue has also received media coverage; see Appendix H.

A brief summary of the main points of the controversy is as follows. AGs may provide additional price competition and therefore lower prices even further. However, the presence of AGs may deter generic entry, which will lower price competition. The effect of generic entry is the key uncertainty in these articles, and this is what our paper aims to shed light on.⁸

3 Data and Descriptive Statistics

We leverage data on quarterly sales and revenue of prescription drugs in the US from 2004-2016, and use this to construct market shares and prices. We define a market to be a molecule-formulation; and of the 241 molecule-formulations we restrict our attention to, about 45% see an AG released. Most AGs are released soon after the first generic entry occurs. Most markets have between 1-11 competitors. We run descriptive regressions to show how market share and prices vary in our dataset, and corroborate the patterns stated in the previous section.

3.1 Data

The data for this paper comes from IQVIA’s National Sales Perspective™(NSP) database and covers sales in US for 2004 - 2016. NSP™ data contains national sales of prescription drugs to pharmacies, clinics, hospitals, and other distribution channels, aggregated to the quarterly level. Each observation includes the name of the molecule (active ingredient) and

⁸There are additional sources of controversy that our paper cannot address. AGs can be used instead of monetary payments to delay generic entry as part of pay-for-delay. AGs can also be used to shield companies from criticisms about high drug prices.

branded name (if relevant), quarter, sales amount in drug unit volume, sales amount in U.S. dollars, supplier of the product, distribution channel, product form (e.g., oral, injectable), and other information including therapeutic class, product launch date, etc.

We define a “drug” to be a group of products having the same molecule (or active ingredient) and product form, following previous literature ([Conti and Berndt \(2020\)](#), [Wang et al. \(2022\)](#), [Yurukoglu et al. \(2017\)](#)). For our purposes, a drug’s therapeutic class describes what broad diagnoses it targets; a drug’s molecule describes the active ingredients present; and a drug’s formulation is its method of delivery, e.g. oral, injectable, etc.

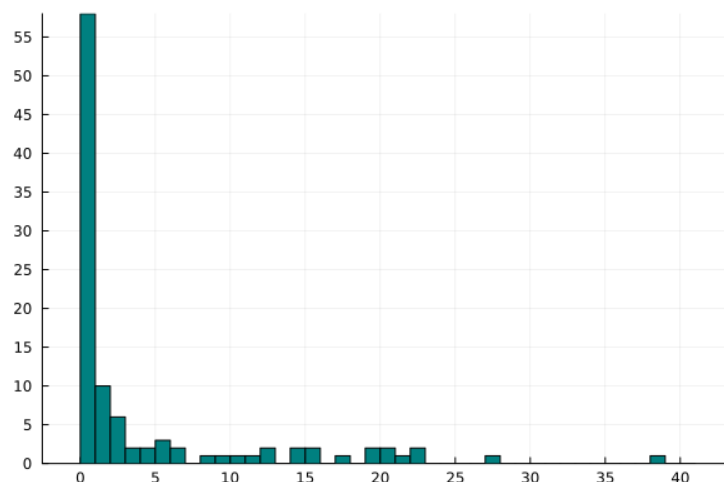
We now give a brief overview of the NSPTM database; this closely follows [Berndt et al. \(2017\)](#), which contains a more detailed description of the entire dataset. The database is constructed through a projected audit that tracks drug sales from manufacturers or wholesalers to pharmacies and other distribution outlets. The measure of “revenue” for each drug is the total amount paid for the purchase of a product to the manufacturer.⁹ Furthermore, the NSP database reports volumes of each drug sold in “standard units”, where a standard unit is defined as the smallest unit of the drug standardized across dosage and therapeutic form ([Dubois and Lasio \(2018\)](#)). We compute prices as the ratio of total revenue to total quantity in standard units, following [Dubois et al. \(2022\)](#) among others. In addition, data on Authorized Generics and Paragraph IV lawsuits were hand-collected.¹⁰ AGs were identified from FDA’s maintained list of AGs, an online source maintained by some producers of AGs, and verifying with news reports on AG release. The list of molecules facing Paragraph IV lawsuits were collected from a publicly available list maintained by the FDA. See Appendix [G.1](#) for links to data sources.

We define the “loss-of-exclusivity” (LOE) period for a molecule-formulation to be the first quarter when any generic appears in that molecule-formulation. While this is restrictive, it suffices for our paper as we are interested in the periods of and following generic

⁹Notably, this measure of revenue does not include rebates that drug manufacturers give back to other intermediaries. Generic manufacturers do not pay rebates, but branded drugs do, so this creates potential measurement error for branded drug prices. To address this, as a robustness check we rerun our demand estimation allowing branded drugs to be charging 30% rebates (following the findings in [Kakani et al. \(2020\)](#) and [Kakani et al. \(2022\)](#)), and obtain very similar results. The presence of brand rebates is used to explain AG release in [FTC \(2011\)](#).

¹⁰We searched news stories about every molecule online to try to ensure there were no non-economic events surrounding them that would affect generic entry. Molecules which faced unnatural events such as lawsuits being settled then revived, facing new lawsuits after entry and being forced to discontinue, facing public pressure such as Epipen, etc. are left out of the final sample.

Figure 2: Histogram of time-difference between first generic entry and AG release period.



This figure is limited to 22 quarters, while in the dataset 4 AGs get released later than 22 quarters. This distribution has a 10th percentile of 0, 25th percentile of 0, median of 0, 75th percentile of 4.25, and 90th percentile of 16.4. See the Appendix for the exact distribution.

entry into a market.

To summarize, for each drug product we see quarterly sales in the US, revenue generated, list of active ingredients, formulation of the drug (oral, injectable, or other), and therapeutic class (ATC3). Sales for a drug-molecule-formulation are aggregated by dosage and strength. Finally, we define a market at the molecule-formulation level.¹¹

¹¹This raises the question of why we define a market at the molecule-formulation level instead of a higher level, such as the therapeutic class. First, we include a molecule-formulation fixed effect in the demand function to account for presence of other molecule-formulations in the same therapeutic class. Second, similar market definitions have been used in various antitrust proceedings and empirical papers ([Conti and Berndt \(2020\)](#), [Starc and Wollmann \(2023\)](#), [Wang et al. \(2022\)](#), [Yurukoglu et al. \(2017\)](#)). Third, the fewer number of products allows us to capture substitution patterns between these bioequivalent products more accurately. Fourth, the branded drug's patent and generic approval holds at molecule-formulation level. Fifth, one drug can be in multiple therapeutic classes which makes modeling therapeutic classes as markets unclear. Finally, there is evidence suggesting that while there can be substitution across molecule-formulations within the same therapeutic class, usually the substitution is from a molecule-formulation without a generic to a molecule-formulation with one ([Arcidiacono et al. \(2013\)](#)). In this data we only look at periods after generic entry, so it is unlikely that people are switching away from the molecule-formulation we are studying.

Table 1: Descriptive statistics

	No. of molecule-formulations
Total	241
AG released	109
Formulation: Oral	162
Formulation: Injectable	58
Formulation: All Others	21
Faced Paragraph IV lawsuit	62

3.2 Descriptive Statistics

While our dataset has comprehensive coverage across all prescription drugs sold in US, we look at a smaller subset of drugs for our project. We select molecule-formulations that saw its first generic entrant after 2004 and before 2016, for which we can identify presence or absence of AG with certainty, and those that are not surrounded by strong external circumstances (e.g. media outrage, serious or repeated lawsuits). This leaves us with 241 molecule-formulations.¹²

Table 1 shows descriptive statistics about our selected molecule-formulations: of the 241 molecule-formulations, about 45% see an AG released, and over 67% are oral formulations. Note that each molecule-formulation has one branded drug and can have at most one AG.

Figure 2 shows when the AGs were released in these markets relative to generic entry. This is a histogram that shows how many quarters after first-generic-entry is AG released. 58 markets see AG release immediately with the first generic entry, 10 markets see AG release one quarter after the first generic entry, and 6 markets see AG released two quarters

¹²We now discuss how this sample selection ties in with our research question and interpretation of results. For studying AG entry, we believe this way of identifying markets AG may enter is quite realistic. AGs are always only released in response to generic entry. In fact, [FTC \(2011\)](#) show company documents where, when outsourcing AG manufacturing, brand manufacturer imposes strong restrictions on the partner AG manufacturers to prevent them from launching the AG before the first generic has entered. Therefore, identifying markets with at least one generic as markets where the AG may enter is realistic, and so our AG cost estimation should be unaffected.

One type of market we do rule out with this is a market where no generic enters, but would have entered had it not been for the threat of AG release. We also rule out markets where generics don't enter because of settlements, profitability concerns, or litigation issues. Most important prescription drugs do face generic entry.

later. AGs continue being released sporadically after the third quarter, with the latest release happening 41 quarters after loss-of-exclusivity.

Next we look at the competitive landscape in our selected markets, specifically regarding generic presence. Most markets have between 1-11 generic competitors, with the maximum number of generics in a molecule-formulation being 19 in our sample. See Figure 5 in the Appendix for the distribution of total generics that enter into a molecule-formulation.

Table 15 in the Appendix has additional descriptive statistics on within-molecule-formulation market shares and revenues, and we summarize some of them here. A generic in a given quarter has a median within-molecule-formulation market share of 18%, while an AG has 22.04%. We also calculate a conservative estimate of lifetime revenue of a product by adding up all the revenue it generated within our dataset range; this is conservative because generics and AGs can continue selling their products after our final available data period. The median AG product has a lifetime revenue of \$54.45 million, the 75th percentile is \$201.78 million, and the 90th percentile is \$444.5 million. More details can be found in Table 15. The magnitude of AG’s lifetime revenues further shows their importance in the prescription pharmaceuticals industry and thus the need to understand their market impact.

Next we study the competitive effects of generic entry on market shares, revenue, and prices; the results are reported in Table 2. Two patterns are particularly notable here. First, the first few generics have a large impact on the brand’s market share and revenue; generics that enter later have a much smaller impact. This suggests that most switching from brand to nonbrand products happen with the first few generics. Second, the brand’s price stays the same or rises as more generics come in. This pattern has been documented previously in the literature.

Finally we look at the generic entry rate in our dataset. We define the “generic entry rate” at any quarter-since-LOE as the number of generic entries happening in that time-frame in our sample, divided by the total number of generic entries observed in our sample (1652). Figure 3 shows the generic entry rate by quarters-since-LOE. The most notable pattern is that a large fraction of the generic entry happens in the quarter when LOE happens (around 33% of all generic entries).

Recall that generics usually apply for FDA approval before LOE; this suggests that a fraction of generics receive approval to launch by the LOE period. However, the majority of generic entry happens after the first quarter. By 20 quarters-since-LOE, about 93% of

Table 2

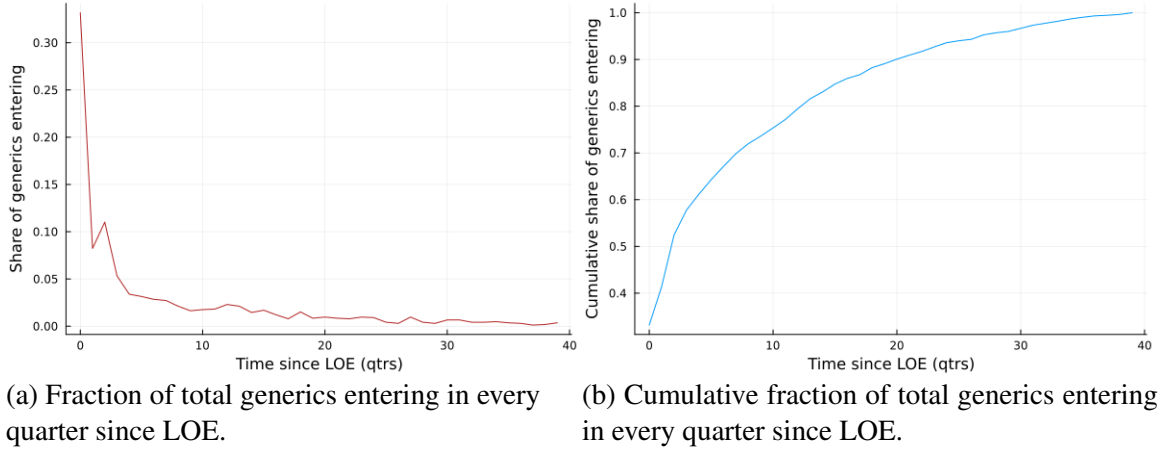
	log(share)	log(price)	log(revenue)
Brand	1.590*** (0.080)	0.291*** (0.029)	1.938*** (0.080)
Brand x qtrs since LOE	-0.057*** (0.003)	0.007*** (0.001)	-0.059*** (0.003)
Authorized generic $\times$ Count of generics	-0.133*** (0.013)	-0.143*** (0.005)	-0.278*** (0.013)
Generic $\times$ Count of generics	-0.216*** (0.008)	-0.138*** (0.003)	-0.358*** (0.008)
Brand $\times \mathbb{1}(\text{count of generics} \leq 2)$ x Count of generics	-0.569*** (0.038)	0.038** (0.014)	-0.502*** (0.038)
Brand $\times \mathbb{1}(\text{count of generics} > 2)$ x (Count of generics - 2)	-0.103*** (0.015)	0.100*** (0.006)	-0.007 (0.015)
Generic $\times$ AG presence	-0.300*** (0.057)	-0.182*** (0.021)	-0.475*** (0.057)
Brand $\times$ AG presence	-0.483*** (0.083)	0.004 (0.031)	-0.497*** (0.083)
Molecule-Formulation Fixed Effects	Yes	Yes	Yes
Obs.	30,822	30,822	30,822

Notes: $\mathbb{1}(\cdot)$ is an indicator for whether the condition inside the parentheses is true. The variable “qtrs since LOE” is the number of quarters since loss-of-exclusivity. “AG presence” is an indicator for whether an authorized generic is present in the market. “Generic” and “Brand” are indicators for generic and brand drugs respectively.

the generic entries have taken place. The spike around 3 quarters-since-LOE reflects some generics entering after waiting out the 180-day exclusivity period of a rival generic.

An alternative model of the industry could have been that the generics all receive final approval as soon as LOE occurs, and the variation in entry times is purely manufacturing frictions that is staggering out product entries over time. This can be ruled out for several reasons. First, [Wang et al. \(2022\)](#) use exact dates from the Orange Book on when a generic receives FDA approval, and identifies that date as “entry”. In their Appendix, they show that about 30% of generics receive FDA approval during the month of LOE, which is ex-

Figure 3: Generic entry rate.



tremely close to the launch pattern we see in our data. They also state: “In the data we observe that around 40% of the market entries occurred within the first year, and more than half of the entries happened within the first two years after the patent expiry.” Therefore the majority of generic approvals in fact happen after the quarter of LOE. Second, generics generally launch as soon as they receive approval because they want to start recouping their sunk cost of entry, and the earlier periods of the market are the most profitable.

4 Model

To study the interaction of generic entry and AG release decisions, we develop a structural model of entry and price competition between AG and generics. The aim is to develop a simple tractable model that captures the important economic mechanisms. On the demand side, we set up a discrete choice model of demand, where the consumer first chooses a group (brand, nonbrand, or outside option) then chooses a product within the group. On the supply side, we construct a two-stage model. The first stage is a static game of generic entry. The second stage is a dynamic game of Authorized Generic release decision. Every period, generics who applied to enter are stochastically approved for entry by the FDA, the branded drug manufacturer decides whether to release an AG, and the market participants compete on prices.

4.1 Demand

We model prescription drug demand as proceeding in two stages. In Stage 1, for a given molecule-formulation, each consumer chooses a group – brand, nonbrand, or outside option (forgoing medication entirely, switching to another molecule, etc.). The non-brand group includes both generics and AG (if present). In Stage 2, each consumer picks a product from the group chosen in Stage 1.

We interpret the decision in Stage 1 as made by a mix of patient, provider, and insurer; and in Stage 2 by pharmacy, procurement officers, or other sellers. This setup matches both institutional reality and data patterns very well.

Stage 1: For molecule-formulation m , let g denote a group, where a group is brand, nonbrand, or outside option. Therefore, $g \in \{\text{brand, non-brand, outside option}\}$.

In Stage 1, the consumer chooses a group. The utility of consumer i for group g in market m at time t is given by:

$$u_{igmt} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} \ln p_{gt} + \beta_1^{(1)} \text{brand}_g + \beta_2^{(1)} \text{brand}_g \cdot \text{time-since-loe}_t + \xi_{gt}^{(1)} + \varepsilon_{igt} \quad (1)$$

where $\ln p_{gt}$ denotes the average of log prices within the group, "brand" is an indicator for the brand group, $\xi_{gt}^{(1)}$ allows for unobserved group-specific demand shocks (such as consumer perceptions), and ε_{igt} is distributed i.i.d. Type-1 Extreme Value. The variable "time-since-loe" measures the number of quarters since generic entry (loss of exclusivity), and this is used to form a brand-specific time trend.¹³ Note that the brand group has a single product, while the non-brand group can have multiple products. Thus $\ln p_{gt}$ is the branded drug's price for the brand group, and is the mean of all non-branded drugs' prices for the non-brand group. The price of the outside option is normalized to zero as is standard in the discrete-choice estimation literature.

¹³The time trend captures two issues. First, branded drug manufacturers can sometimes take actions that slow down generic adoption. This involves things like product hopping, getting approval for new therapeutic uses, gaining orphan drug designation, etc. These moves stop consumers and pharmacists from switching over to generic drugs and get around automatic substitution laws. While our model does not specifically account for such actions, our demand specification contains a brand-specific time trend that aims to capture the idea that the branded drug loses its appeal over time as the manufacturer runs out of such tactics. As would be expected, we find that the branded drug becomes less attractive over time, possibly because such tactics become less effective over time. Second, this could also capture new drugs being discovered that cause people to switch to different molecule-formulations.

Stage 2: If the consumer chose the non-brand group in Stage 1, and there are multiple non-brand drugs available for the molecule formulation m at time t , then the patient has to choose from among these products.

In Stage 2, the consumer chooses a specific non-brand product. The utility of consumer i choosing non-brand product j in market m at time t is given by:

$$u_{ijt} = \gamma_{m(j)}^{(2)} + \alpha^{(2)} \ln p_{jt} + \beta_1^{(2)} AG_j + \xi_{jt}^{(2)} + \varepsilon_{ijt} \quad (2)$$

where $\ln p_{jt}$ denotes the log price of product j , AG_j is a dummy for product j being an Authorized Generic, $\xi_{jt}^{(2)}$ allows for unobserved product-specific quality, and ε_{ijt} is distributed i.i.d. Type-1 Extreme Value.

Given our assumptions, the market share of a single good can be expressed as:

$$s_j = s_{g(j)} s_{j|g(j)} \quad (3)$$

where s_g is the probability of the consumer choosing group g , and $s_{j|g}$ is the probability of the consumer choosing product j conditional on group g . Note that the “brand” group and “outside option” group have a single product in each group, and so $s_{j|g(j)} = 1$ for these groups.¹⁴

4.1.1 Discussion on demand model

While we refer to a single “consumer” throughout, we interpret different agents making decisions in each stage. We regard Stage 1 as representing a decision made by the patient, physician, and insurer, while Stage 2 represents a decision made by pharmacies, procurement officers at the provider, and other sellers.

This two-stage model has two advantages. First, it reflects institutional reality where the patient/provider make the decision on which molecule-formulation to purchase and

¹⁴Stage 1 gives the expression for the market share of a group, while Stage 2 gives the expression for the market share of a product conditional on a group. The probability of choosing group g , s_g , is given by the logit formula $s_g = \frac{e^{\delta_g^{(1)}}}{\sum_l e^{\delta_l^{(1)}}}$ where $\delta_g^{(1)} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} \ln p_g + \beta_1^{(1)} \text{brand}_g + \beta_2^{(1)} \text{brand}_g \cdot \text{time-since-loe} + \xi_g^{(1)}$. Conditional on choosing nonbrand group, the probability of a consumer choosing product j from nonbrand group (denoted by $s_{j|g}$) is given by $s_{j|g} = \frac{e^{\delta_j^{(2)}}}{\sum_k e^{\delta_k^{(2)}}}$ where $\delta_k^{(2)} = \gamma_{m(k)}^{(2)} + \beta^{(2)} AG_k + \alpha^{(2)} \ln(p_k) + \xi_k^{(2)}$.

whether to buy the branded product, then the pharmacist or procurement officer decides which specific product to supply to the patient. (For instance, pharmacies choose which generics to stock, as described in [Cuddy \(2020\)](#).) Second, for estimation this matches our data patterns better than other traditional discrete-choice specifications.

See Appendix [B.1](#) for an in-depth discussion about contrasting our model with traditional discrete-choice specifications. See Appendix [B.3](#) for results from random-coefficients discrete choice estimation, which often provide unintuitive parameter values or do not match the data patterns well.

While our demand model is somewhat different from other discrete-choice settings, note that our aim of the project is to study entry decisions; as long as we correctly predict prices and per-period profits, our central arguments work.¹⁵

4.2 Supply

We first present an overview of our supply model. The following subsections explain the components of the model, and Section [4.3](#) goes over the reasoning behind the modeling choices.

Recall that generic manufacturers are slowed down from entering due to regulatory delays; as a result, they apply for entry well in advance of LOE. In contrast, AGs can enter at any time. This motivates our model setup below.

There are two stages to the supply side:

1. Stage 1: Generic firms decide whether to apply to enter a market or not. This stage is modeled as a game of simultaneous entry application between identical firms. The generics are forward-looking and form rational expectations about AG entry and regulatory delays when making entry decisions.
2. Stage 2: Loss-of-exclusivity happens and a dynamic game begins. The dynamic game lasts T periods. Every period, a stochastic number of the generic firms which applied for entry are approved and enter into the market. The branded drug manufacturer decides whether to release AG or not (where the AG release decision is irreversible). Then, all the firms set prices and receive payoffs.

¹⁵Another option is to use a flexible regression on observed market shares or revenues to predict firm revenues in different states; an example of such a regression is presented in Table [2](#).

We now explain the model in reverse, starting from payoffs in the second stage.

4.2.1 Second-stage

The branded drug manufacturer's per-period payoff is:

$$\pi^b(s_{mt}) = [P_{mt}^b - MC_m^b]s_b(s_{mt})M_m + \mathbf{1}(AG_{mt} = 1) \left[[P_{mt}^{AG} - MC_m^{AG}]s_{AG}(s_{mt})M_m \right] \quad (4)$$

where s_{mt} denotes the state variable vector in market m , superscripts and subscripts b and AG refer to the branded product and AG respectively, P denotes price, MC denotes marginal cost, $s(s_{mt})$ denotes market share, and M_m denotes market size. The indicator variable $\mathbf{1}(AG_{mt} = 1)$ denotes whether an AG has been released onto the market. The intuition behind the branded drug manufacturer's payoff is that it makes the first half of the payoff from the branded drug, and the second half from the AG conditional on releasing the AG.

Similarly, the generic firm l 's per-period payoff with the firm-specific state variable vector $s_{l,t}$ is:

$$\pi^g(s_{lmt}) = (P_{mt}^g - MC_m^g)s_g(s_{lmt})M_m \quad (5)$$

The branded drug, generics and AG compete on prices. We use a reduced-form price prediction regression to predict prices in different states and markets; see Section 5.2. While the model has been written for maximum generality, during estimation we set $MC_b = MC_{AG} = MC_g$ for a given market m . This assumption can be relaxed.

A dynamic game begins from $t = 1$ and lasts T periods, where every period is a quarter. Let n_e^* be the number of generic firms that have applied for entry (which is determined in the first stage). In period $t = 2$ the branded drug's market exclusivity expires, and every period a random number of generic firms gain FDA approval and enter the market.

Since AG release is a dynamic decision, the branded drug manufacturer's choice problem can be written in terms of value functions. Thus, the value function for a branded drug manufacturer every period is given by:

$$V^b(s_{mt}, \epsilon_{mt}) = \max_{AG_{mt+1} \in \{0,1\}} \pi^b(s_{mt}) - \mathbf{1}(AG_{mt} = 0, AG_{mt+1} = 1) \kappa_m^{AG} + \beta E[V^b(s_{mt+1}, \epsilon_{mt+1}) | s_{mt}, \epsilon_{mt}] + \epsilon_{mt}(AG_{mt+1}) \quad (6)$$

where $AG_{mt+1} = 1$ means the AG is in the market m during period $t + 1$. AG entry cost in molecule-formulation m is denoted as κ_m^{AG} . The AG entry decision is irreversible, i.e. $AG_{mt} = 1$ implies $AG_{mt+k} = 1 \forall k > 0$. AG entry is implemented with a one-period delay.

The timing of AG release could be influenced by factors observed to the manufacturer but unobserved to the econometrician. To account for such factors, the model (following the existing literature on dynamic games) includes choice-specific payoff shocks $\epsilon_{mt}(AG_{mt+1})$. These shocks follow a Type 1 Extreme Value distribution.

The probability of releasing an AG conditional on not having been released previously is¹⁶:

$$Pr(AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 0) = \exp\left(\frac{-\kappa_m^{AG} + E[V^b(s_{mt+1}, AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 0)]}{\sigma_\epsilon}\right) / B \quad (7)$$

where $B = \exp(-\kappa_m^{AG} + E[V^b(s_{mt+1}, AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 0)]) + \exp(E[V^b(s_{mt+1}, AG_{mt+1} = 0 | s_{mt}, AG_{mt} = 0)])$.

The probability of releasing an AG conditional on having released AG in the past is one, i.e. AG release is an irreversible decision:

$$Pr(AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 1) = 1 \quad (8)$$

The value function for generic l is given by:

$$V^g(s_{lmt}) = \pi^g(s_{lmt}) + \beta E[V^g(s_{lmt+1}) | s_{lmt}] \quad (9)$$

where $s_{l,t}$ includes whether generic l has been approved for production by the FDA. While the generic manufacturers do not face a dynamic decision in the second stage, they use the value function to make entry decision in the first stage.

¹⁶This expression involves a slight abuse of notation: s_{mt} already includes the value of AG_{mt} , but we separate them out in the arguments here for clarity.

From period T , all manufacturers make zero profits, i.e. $\pi^b(s_{mt}) = \pi^g(s_{mt}) = 0$ for $t \geq T$. This accounts for small economic profits in mature drug markets due to new drug discovery, patent expiration of other branded drugs, and the extremely-low profit margins generally observed in a mature generic drug market.¹⁷

The manufacturers have to form expectations over generics' entry probabilities; recall that entry happens stochastically over time due to regulatory delays. To express the value functions, we need to pin down how state transitions happen regarding generic entry. Thus, we model the evolution of number of generics present as a binomial process following Ching (2010).

Suppose by period t in market m , $\mathcal{N}_{mt}$ generic entrants have already received approval and launched their products. Thus $n_{em}^* - \mathcal{N}_{mt}$ entrants have not yet received approval. The probability that of the $n_{em}^* - \mathcal{N}_{mt}$ remaining entrants, k will receive approval in this period is given by:

$$P_e(k, \mathcal{N}_{mt}, t) = \binom{n_{em}^* - \mathcal{N}_{mt}}{k} \lambda(t)^k (1 - \lambda(t))^{n_{em}^* - \mathcal{N}_{mt} - k} \quad (10)$$

where $\lambda(t)$ – the probability of a single generic entering the market – is estimated from the data. We assume the branded drug manufacturer knows n_{em}^* and total generics that enter for $t = 2$ ($\mathcal{N}_{m2}$).

4.2.2 First-stage

In the first stage, a large number of generics decide whether they want to enter. We assume all generic firms are ex-ante identical, do not receive private error draws for entering and staying out, and do not know their draws of ξ_{jt} conditional on entry.

Generics keep applying for entry until the value from entering no longer exceeds the cost of entry. More precisely, the equilibrium number of generic entrants is n_e^* , determined by the following inequality:

$$V^g(s_{gm0}, n_{em}^*) \geq \kappa_m^g > V^g(s_{gm0}, n_{em}^* + 1) \quad (11)$$

where κ_m^g denotes the generic manufacturer's entry cost for molecule-formulation m and

¹⁷This assumption can be relaxed. For instance, after period T , the market state could be set at s_{mT} (the state reached in period T), with the manufacturer receiving the corresponding payoff for infinite periods: $V^b(s_{mT}) = \sum_{\tau=T}^{\infty} \beta^\tau \pi^b(s_{mT})$ for brands, $V^g(s_{mT}) = \sum_{\tau=T}^{\infty} \beta^\tau \pi^g(s_{mT})$ for generics.

s_{gm0} is the initial state of the market from a generic’s perspective (i.e. before LOE and generic entry have begun). Intuitively, the value function of entry is declining in the number of entrants – more entrants mean more competition in the market and lower per-period profits. At the equilibrium n_{em}^* , current entrants can cover their entry cost, but an additional entrant would stop that from happening. This follows the free-entry condition in [Igami \(2018\)](#). Finally, unlike the branded drug manufacturer, the generics do not know $\mathcal{N}_{m2}$.

4.2.3 Model solution

We solve the model by backward induction and recover the equilibrium number of generic entrants and policy function of AGs. Since the AG’s policy function enters the value functions of generic entrants and the AG knows n_{em}^* , all agents are forming rational expectations about each others’ choices when making a decision. This means that a policy or environment change that affects the choice probabilities of one agent will have feedback effect on other agents in the market. Our model solution adapts the method of [Igami \(2018\)](#); details of the model solution are given in [Appendix A.1](#).

4.3 Discussion

Interpreting AG’s choice-specific shocks: Recall that the AG release decision faces choice-specific payoff shocks $\varepsilon_{mt}(AG_{t+1})$ that follow a Type 1 Extreme Value distribution. These logit shocks have a logit scaling parameter σ_ε that governs the variance of these shocks. Intuitively, this is used to account for payoff shocks observed to the agent but unobserved to the econometrician. These payoff-shocks could subsume a variety of factors such as unobserved constraints, manufacturing delays, cost issues, policy/legal uncertainty, inexperience with generic production, organizational/transactional frictions, etc. In our setting, it also subsumes cases where an AG was not released due to a No-AG settlement.¹⁸ This paper does not model No-AG settlements explicitly, partly because it is extremely difficult to identify when such settlements have happened. Our demand model is not affected by No-AG settlements, and our counterfactuals apply to markets where such settlements have

¹⁸“No-AG settlements” are settlements stemming from Paragraph IV certifications where the first-filer generic firm agrees to delay entry in exchange for assurance that an AG will not be released. Such a No-AG settlement could be related to a Paragraph IV settlement regarding the same molecule-formulation, or a settlement involving a different molecule-formulation but involving the same firms.

not occurred. See [FTC \(2011\)](#) and [Bokhari et al. \(2020\)](#) for a more detailed analysis. Finally, the choice-specific payoff shock subsumes the brand manufacturer engaging in other post-LOE strategies (such as product line extension or product discontinuations) that affect its AG release decision. See Appendix [D](#) for a detailed discussion on factors subsumed in these choice-specific shocks.

Discussion on price prediction equation: Instead of assuming a pricing conduct such as Nash-Bertrand or Nash-in-Nash bargaining, we use a price prediction regression to predict prices charged by different firms in different markets. Any specific pricing conduct assumption is unlikely to be very accurate, since pricing in the pharmaceutical industry is highly complicated.¹⁹ Instead, our price prediction regression is a flexible way to model the pricing game without making strong assumptions about the pricing conduct. The advantage is that our predicted per-period profits are not susceptible to misspecifications about the pricing conduct assumption.²⁰

Note that in our counterfactuals, we do not change any primitives that would affect the static pricing game. Otherwise, the price prediction equation estimated from the observed data would no longer be a good predictor, as a change in demand primitives should change the underlying pricing patterns. Instead, we limit our counterfactuals to changing parameters that affect dynamic incentives (such as entry restrictions and faster generic entry rates).

Our price prediction equation is similar in spirit to the price-setting equation in [Maini and Pammolli \(2023\)](#). In their paper, the authors try to predict prices across different European countries with different bargaining policies and unknown objective functions. In both our paper and [Maini and Pammolli \(2023\)](#), the main focus is on entry instead of the static pricing game, which means being able to accurately predict per-period payoffs is sufficient

¹⁹Other papers which study pharmaceuticals with aggregate data and make a pricing conduct assumption include [Arcidiacono et al. \(2013\)](#), [Dubois et al. \(2022\)](#), and [Starc and Wollmann \(2023\)](#). Unlike these papers there are major differences in our analysis that preclude us from doing the same. [Arcidiacono et al. \(2013\)](#) focus on a small number of molecules, which allows them to leverage additional data and impute rebates. In contrast, our paper considers well over 200 molecule-formulations. [Dubois et al. \(2022\)](#) restrict their attention to sales in the hospital channel for drug sales, while we consider all channels. They assume Nash-Bertrand, which is a more reasonable assumption in their setting. [Starc and Wollmann \(2023\)](#) consider the pharmacy channel and use different datasets compared to this paper.

²⁰A regression of prices on level of competition may suffer from endogeneity issues from demand shocks. This is less of an issue than in other settings, because generics cannot simply respond to a demand shock and enter the market - they need to wait years to get FDA approval for marketing. Furthermore, we also add in molecule and formulation fixed effects.

for our analysis.

Another issue is that during this period, some generic manufacturers may have been engaging in price-fixing (Cuddy (2020), Starc and Wollmann (2023)). This occurred from late 2013 onwards; our dataset’s coverage is from 2004-2016. As a result, we rerun the price prediction regression by dropping years 2013 onwards, and obtain very similar results.

Generics’ entry timing and Paragraph IV certification: Recall that we model the entry decision by generics to be a static entry game that happens before loss-of-exclusivity. We choose to model generic entry as a static game instead of a dynamic timing game because regulatory delays create uncertainty for generics about when they can enter the market; as a result, generics usually apply for FDA approval well in advance of the LOE.

Because nearly all generics apply for entry before LOE, a fraction get final approval from the FDA as soon as LOE happens (over 30% in our data). This is accounted for in the generic entry rate calculation; see Section 5.3 for a detailed explanation. The concern that some generics file for entry well after loss-of-exclusivity turns out not to be much of an issue for our setting. Among others, Starc and Wollmann (2023) find that the mean ANDA filing in mature generic markets is 0.21 per drug-year (even lower for non-cartelized markets), and conclude that “entry into (mature) generic markets is typically rare”.

One simplification we make is regarding generics’ decision to file a Paragraph IV certification while applying for entry and any subsequent pay-for-delay settlements.²¹ The decisions to file Paragraph IV certification and negotiate any settlements (such as pay-for-delays or no-AGs) are exogenous to our model. Instead our model assumes the following: each generic assigns some probability that there is a Paragraph IV certification, that they are the first-filer, or that they have to wait 180 days to enter because a rival achieved the first-filer status instead. These probabilities are reflected in our calculation of generic entry rates. The probability of a no-AG settlement happening is subsumed in the variance of the AG’s choice-specific shocks. Thus, we think of Paragraph-IV certification and first-filer status as being assigned randomly to one of the entrants outside the model. Furthermore, we also have to abstract away from pay-for-delay settlements and the negotiations surrounding them. Our model ignores how pay-for-delay incentives affect entry application

²¹When a generic files a Paragraph IV certification, the branded drug manufacturer can agree to pay them directly to abandon the patent challenge or delay entry as a first-filer. Such settlements are called pay-for-delay settlements.

decision, and instead focuses on the periods of and following generic entry.²² These are simplifications driven by data limitations and computational concerns, and they allow us to tractably model the generic entry decision.

Information assumptions: We discuss our assumptions on information regarding the number of generic entrants. First, we assume that the AG knows how many generics have applied for entry to its molecule-formulation. In reality, the information on ANDA application by a generic manufacturer becomes public only after it wins ANDA approval from the FDA (although generic manufacturers may have private information sources that could alert them to ANDA filings by rivals). This is a simplifying assumption that improves the tractability of our model. Second, we assume that the AG correctly predicts the number of generic launches in the quarter when loss-of-exclusivity happens; this is realistic as the FDA would have announced before loss-of-exclusivity which ANDAs have been approved by then.

5 Estimation and Results

We use a reduced-form price regression to predict prices, and estimate generic entry rates from the data. Stages 1 and 2 of the demand model is estimated separately using the logit-inversion techniques from [Berry \(1994\)](#) and [Maggio et al. \(2022\)](#) respectively. Marginal costs are estimated by matching predicted generic entry numbers with observed generic entry numbers. AG release decision’s logit scaling parameter (that governs the choice-specific payoff shocks) is estimated via Maximum Likelihood.

²²The thought experiment could be something as follows. The generic manufacturer has two divisions: the first decides whether to enter a molecule formulation, the second division decides whether to file a Paragraph IV certification, then negotiate for a pay-for-delay settlement or no-AG settlement. Our paper could be thought of as modeling the actions of the first division, while the second division adds some stochastic payoff/loss to the overall profit based on how the Paragraph IV/settlement negotiations work out. With detailed data on settlement outcomes it is possible to model the actions of the second division as well. This thought experiment is of course a simplification of the very complicated process of generic entry and settlement negotiations that happens in reality.

5.1 Heterogeneity by channel presence

Our dataset contains a breakdown of drug sales by the channels where the sale happened. These channels are retail pharmacies, clinics, federal facilities, food-stores, long term care hospital, HMO, home health care, mail services, and non-federal facilities (Berndt et al. (2017), Dubois and Majewska (2022)). In our analysis we want to be cognizant of the fact that demand parameters and pricing behavior can vary across different distribution channels (e.g. pharmacies versus hospitals).

To that end, we classify all channels into one of two types: patient-facing or non-patient-facing. “Patient-facing channels” are channels where the individual patient has more influence over group choice, whereas “non-patient facing channels” are where patients have limited influence (e.g. the group is chosen by the hospital where the patient is admitted). The channels we classify as “patient-facing” are retail pharmacies, mail services, and food stores; the remaining channels are classified as “non-patient facing”.

A molecule-formulation is classified as patient-facing if more than 50% of its sales are in patient-facing channels. In our data, we classify 183 molecule-formulations as patient-facing and 58 molecule-formulations as non-patient-facing. The demand estimation and price prediction regression are estimated separately for patient-facing and non-patient-facing molecule-formulations.

5.2 Price prediction and generic entry rates

Instead of assuming a pricing conduct (such as Nash-Bertrand or Nash-in-Nash bargaining), we choose to be agnostic about the price-setting behavior and instead predict prices using observables. More specifically, we regress observed prices on various covariates, and this regression is used to predict prices in different industry states. We use the regression reported in Table 3 for this purpose.

We make several simplifying assumptions. First, we assume that all non-brand drugs charge the same price in a market-quarter.²³ Second, we leave out quarter fixed effects in our demand system, and instead add the median estimated quarter fixed effect to the demand system’s intercept term.

²³In general, generics price very close to each other in the market. Furthermore, in our data we see AGs price close to other generic manufacturers in the market, a fact also corroborated in FTC (2011).

Table 3: Price prediction regressions

	Patient-facing	Non-patient-facing
formulation: Injectable	2.118 (0.136)	– (–)
formulation: Oral	-1.308 (0.112)	– (–)
No. of nonbrand drugs	-0.269 (0.009)	-0.298 (0.016)
No. of nonbrand drugs squared	0.008 (4.843e-04)	0.010 (0.001)
Brand	0.123 (0.031)	-0.076 (0.041)
Brand * time-since-generic-entry	0.017 (0.001)	-0.005 (0.002)
Brand * No. of nonbrand drugs	0.232 (0.004)	0.231 (0.008)
Brand * Authorized generic present	0.041 (0.031)	0.774 (0.064)

Notes: We regress log of prices on the covariates listed above. Robust standard errors are in parentheses.

We combine this price prediction equation and estimated marginal costs with our estimated market share equation (Equation (3)) to generate per-period profits in different states.

5.3 Estimating generic entry rates

To operationalize equation (10), we estimate the average generic entry rates $\lambda(t)$ at every quarter-since-LOE. This is done as follows. For each molecule-formulation and quarter-since-LOE, we calculate the number of generics waiting to enter, i.e. final number of generics observed in that molecule-formulation minus the current number of generics present in the molecule-formulation. We sum this number across all molecule-formulations at a given quarter-since-LOE to get the remaining entrants for every quarter since loss-of-exclusivity in our dataset. Then we find the total generic entry that happens in every each molecule-formulation and quarter since loss-of-exclusivity.²⁴ We sum this number across all molecule-formulations at a given quarter-since-LOE to get the total generic entry for every quarter-since-LOE in our dataset. Taking the ratio of total generic entry by remaining entrants at every quarter-since-LOE gives us the average generic launch rates at every quarter-since-LOE.²⁵

We argue this measure of generic entry rates is correct for our setting. First, this is the correct empirical probability distribution that generics use to form beliefs about their entry timing. Second, regulatory delays are subsumed in this measure. Therefore, if regulatory delays were reduced, that can be interpreted as the generic entry rates at each period being higher.

Nearly all generics apply for FDA approval well in advance of loss-of-exclusivity, and some receive it before or as soon as LOE happens. This is why we see over 30% of the generic entries happening in the quarter of LOE. This also explains why we find generic

²⁴A generic manufacturer’s entry into a molecule-formulation is identified as the earliest quarter that manufacturer is spotted in our dataset for that molecule-formulation.

²⁵In our calculation of generic entry rates, we limit the number of entrants to those happening on or before 17 quarters. Our main worry is that entry happening after 17 quarters could involve generics who applied well after loss-of-exclusivity; filings happening after LOE could be for other reasons not modeled in our framework. [Starc and Wollmann \(2023\)](#) find that the majority of ANDA filings are approved within 4 years of filing, hence we limit our analysis to 17 quarters, and robustness checks with 24 quarters are very similar. It turns out to matter little for our results because there are few generic entries happening after 17/24 quarters in any case.

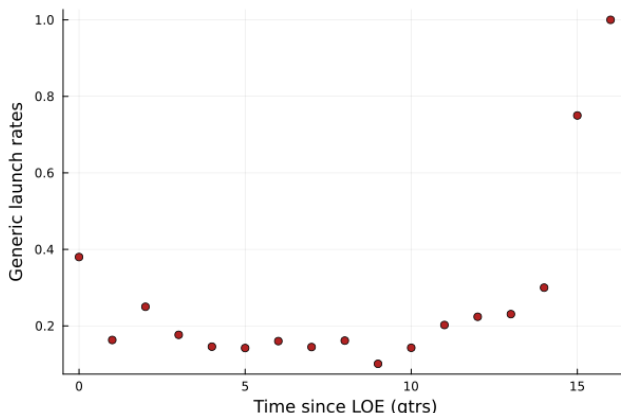


Figure 4: Generic entry rate

entry rates to be highest at the beginning of loss-of-exclusivity, and declining over time. We also implicitly assume that once enough time has passed the generic entry rate rises up to 1.0, which imposes that all generic applicants eventually enter. This rules out cases where i) a generic applies for entry and never receives approval and ii) a generic receives approval but chooses to never enter.

5.4 Estimation of demand-side parameters

We estimate each stage of the demand model separately. Stage 1 is estimated following the logit inversion of [Berry \(1994\)](#). We run the logit share regression on group-level market shares, and use Gandhi-Houde instrumental variables ([Gandhi and Houde \(2019\)](#)). Stage 2 of the demand model is a multinomial logit demand model but without an outside option. Following [Maggio et al. \(2022\)](#) this can be estimated using a linear regression of logs of market share on market-by-quarter fixed effects and other covariates. We allow our demand parameters to vary based on whether the molecule-formulation is “patient-facing” or “non-patient facing”. See [Appendix B.2](#) for a detailed explanation of our demand estimation strategy.

5.5 Estimation of supply-side parameters

We estimate the supply-side cost parameters of our model. First we estimate marginal costs by molecule-formulation by matching predicted generic entry numbers to observed numbers at publicly reported generic entry costs. Next, we estimate the logit scaling parameter of the AG choice-specific shocks via Maximum Likelihood. Throughout we set the quarterly discount factor at $\beta = 0.95$.²⁶

First we derive approximate bounds on the marginal costs by matching predicted generic entry numbers to observed numbers at publicly reported generic entry costs for markets with no AGs. Recall that we do not make assumptions about pricing conduct such as Nash-Bertrand to avoid misspecification; as a result, we cannot impute marginal costs from observed prices (though as we will see later, we do use the average observed prices in the molecule-formulation to inform the marginal costs of that molecule-formulation). Instead, our aim is to instead estimate marginal costs from observed generic entry numbers. The intuition is that the estimated marginal costs will scale down predicted profits so that at the publicly-reported generic entry costs, the model predicts reasonable generic entry numbers.²⁷

The estimation is done as follows. We assume that the marginal cost for molecule-formulation m can be approximated by $MC_m = \vartheta \bar{p}_{gm}$, where $\bar{p}_{gm}$ is the average observed price of generics in molecule-formulation m over all periods in our dataset, and ϑ is a scaling parameter. The gain from scaling the average price of generics to estimate marginal costs is that i) generics price close to the marginal cost, so there is information about marginal cost levels from observing their prices²⁸, and ii) it allows marginal cost to vary

²⁶This corresponds to an annual discount factor of 0.81, which is in the range of [0.75,0.95] used by [Igami and Sugaya \(2022\)](#). Some dynamics papers use higher annual discount factor such as 0.95. We choose a lower number because i) we do not account for exit or discontinuation of manufacturers, so the low discount factor may indirectly account for that possibility by downweighting future payoffs more, ii) pharmaceutical industry sees new molecule-formulations experiencing loss-of-exclusivity regularly so there’s a high risk of a product market becoming obsolete in the future. Such risks of product obsolescence is implicitly captured by a lower discount factor.

²⁷This suggests that our interpretation of marginal costs is somewhat broader than the cost of producing an additional pill. Our estimates of marginal costs of operation reflects the cost of operating the factory and production line, maintaining supply chains, transaction costs, etc. It is also the economic cost of launching a product, and so reflects costs beyond accounting costs. For instance, there could be opportunity cost tied to allocating inputs to this product as opposed to elsewhere, or there could be additional constraints facing the managers.

²⁸For example, [Grennan et al. \(2021\)](#) set marginal cost as 0.17 times generic price.

across molecule-formulations. The alternative of having the same marginal cost across all molecule-formulations would be too restrictive, and trying to estimate marginal cost by molecule-formulation is likely to be noisy and computationally costly.

We isolate molecule-formulations for which AGs do not enter, and assume that AGs had zero probability of entering these markets. For a given generic entry cost κ_g , we try different trial values of ϑ and solve for the equilibrium generic entrants in each market. Using grid-search, we pick the ϑ that minimizes the sum of squared differences between the observed and predicted generic entry numbers. Recall that reported generic entry costs vary between \$3 million and \$15 million; therefore we set κ_g to each of these bounds and recalculate ϑ . We find that ϑ range from 0.54 to around 0.65. Thus, we fix per-period fixed costs to be the average of these two bounds for our main estimation. Details of the estimation procedure are given in Appendix A.3.

We also assume for our AG cost estimation and counterfactuals that all manufacturers in a molecule-formulation have the same marginal cost, i.e. $MC_m^b = MC_m^g = MC_m^{AG} = MC_m$.

Next we estimate the logit scaling parameter of the AGs' choice-specific shocks σ_ε and set κ_m^{AG} to zero. We set $T = 32$ (meaning that the dynamic game lasts 32 quarters) and impose that profits become zero afterwards (both assumptions can be relaxed). The parameter is estimated via Maximum Likelihood Estimation; see Appendix A.2 for details.

This raises the question of why we set κ_m^{AG} to zero. First, the generic's sunk cost of entry mostly comes from having to reverse-engineer the production process of the branded drug and setting up the supply chains; this is not a cost the AG release has to incur. Other costs such as production, negotiation, and distribution are captured in the estimated marginal costs. Second, we conduct robustness checks by setting κ_m^{AG} to vary between 0 to \$3 million (which is the lower bound of generic entry costs) and find the estimated σ_ε to be very similar across all these values.²⁹

5.6 Results

Tables 4 and 5 show the results of the demand estimation; we see expected signs on the coefficients. Results of estimation of the logit scaling parameter of AGs are reported in Table 6.

²⁹We also attempted to estimate κ_m^{AG} and σ_ε jointly, but found that they were poorly identified in practice.

Table 4: Results of demand estimation for Stage 1.

	Patient-facing	Non-patient-facing
ln(price)	-1.004 (0.033)	-2.781 (0.244)
Brand	0.799 (0.069)	1.033 (0.250)
Brand * time-since-first-generic	-0.095 (0.007)	-0.008 (0.014)

Notes: Robust standard errors are reported in parentheses. Market and quarter fixed effects are suppressed.

Table 5: Results of demand estimation for Stage 2.

	Patient-facing	Non-patient-facing
Authorized Generic	0.857 (0.077)	0.067 (0.186)
ln(price)	-0.647 (0.023)	-1.286 (0.068)

Notes: The demand model is estimated using a fixed-effects regression following [Maggio et al. \(2022\)](#). Robust standard errors are reported in parentheses.

Table 6: Results of cost estimation.

σ_{ε}	586497.139 (17277.888)
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Notes: Estimation done using Maximum Likelihood. Standard errors are reported in parentheses.

6 Counterfactuals

In this section we first explain how the counterfactuals are conducted, then discuss the results.

To understand our model and predict the impact of different policies, we solve the model by backward induction. For the counterfactuals, we solve for both the number of generic entrants and policy function of AG. A description of the full-solution method is given in Appendix [A.1](#).

Next we explain how we conduct and report our counterfactuals. For each counterfactual we solve the model, run 200 simulations, then report the average results from the simulations. We run the counterfactuals for 200 of the 241 market-specifications (molecule-formulations) found in our data, and study the distribution of effects. Note that we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values. We vary generic entry costs across market-specifications to ensure that the predicted number of generics is not below 1 and not greater than the maximum number observed in the data; more details of the implementation over market-specifications can be found in Appendix [C](#).

We now present a brief discussion on profitability in different phases of a market’s lifecycle. The early quarters of a molecule-formulation after loss-of-exclusivity are the most profitable for firms. During these periods, there are few firms in the market, resulting in high markups and high economic profits. As time goes by, more generics enter the market, resulting in markups being driven close to zero and low economic profits. This provides another reason why generics want to launch earlier – the earlier phase of the market yield the highest profits and so contribute most towards covering the entry cost.

6.1 Deterrence through early-move versus early-launch

In this section, we illustrate how both the independent generics and AG can deter each other’s entry, but in different ways. We find that generics can deter AG release because they make entry decisions earlier; on the other hand, the AG can deter generic entry because it is able to launch earlier and provide price competition from the early stages of the market.

An independent generic has an early-move advantage. This is because a generic makes its entry application decision and incurs its fixed cost in the first stage, before the AG. This

ability to move earlier means that the generic can deter an AG from entering. Note that the AG having a substantially lower entry cost than independent generics (conditional on drawing a trade shock) means that this entry deterrence effect is light.

On the other hand, the AG can deter the independent generic's entry application through the fact that it can launch earlier. Since the AG can launch its product early – and during the most profitable phase of the market – it has a strong entry deterrence effect on the independent generic.

To showcase these economic forces, we perform the following two counterfactuals.

First, we show that the generics are able to deter AG release when AGs do not enjoy the launch-timing advantage over generics. We fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. Thus, in this counterfactual, the AG no longer has the early-launch advantage or the cost advantage. We find that across nearly all market-specifications, the AG is not launched. The results are summarized in Table 8.

Next, we show that AG is able to deter generics better when it has a stronger early-launch advantage. To do so, we vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG always launches in the period of LOE. Next we run simulations where the AG always launches 8 quarters after LOE. We find that for over 80% of market-specifications, the AG can deter more generics when it can launch earlier; the remaining market-specifications see no change in number of generics. The results are summarized in Table 9.

6.2 Generic entry rates

We examine the effects of reducing regulatory delays affecting generic entry. In our counterfactuals, we scale up the probability of generics entering in every quarter after loss of exclusivity, and simulate the impact on market outcomes. That is, reducing regulatory delays corresponds to faster generic entry rates in our model.

We find that faster generic entry rates leads to fewer generics applying for entry, prices

being lower in initial periods, prices being weakly higher in later periods.

Under faster generic entry rates, there are opposing forces on a generic's entry decision. On the one hand, faster entry rates means the generic can market their drug earlier, and so has a longer time (till the terminal period) to make profits and cover their entry cost. This force would incentivize more generics to enter. On the other hand, faster entry rates means that when a generic does enter the market, more of its competitors are already in the market, so it will face greater competition and smaller profits upon entry. This force would disincentivize more generics from entering.

So which of these forces dominate? This is an open question we can study by simulating the market under faster generic entry rates.

For the first group of counterfactuals we leave AGs out of the market for clarity. For most market specifications we see that as generic entry rates increase, there are fewer generic entrants; for the remainder, there is no change in total generics. This means that the entry disincentive from earlier competition generally dominates the entry incentive from longer operation time in the industry.³⁰ The results are summarized in Table 10. An example for a single molecule-formulation is given in Table 7.

Our counterfactual simulations help further illustrate why the second force dominates. Take a molecule-formulation where, if generics can enter immediately, then there will be 3 generic entrants. If an additional generic were to enter, the prices will be driven down so low that the additional generic will not be able to cover its entry cost. Now let's take the same market, but with a slower entry rate. The first 3 generics will still choose to enter, but their entries are now more spaced out over time. What about an additional fourth entrant? If the fourth entrant enters, then in the long-term the prices will be very low as in the immediate case. But in the short term, the entrant might get a lucky early approval and enter the market in its early periods when there's little competition, and make large profits.

³⁰To see why, we can look at the per-period prices (and profits) in the industry. The earliest periods are the most profitable, because there are few nonbrand products in the market, resulting in high markups. As more generics enter, these markups are driven down to zero, so that in the long run price is nearly equal to marginal cost. This means that after a certain number of periods, generic firms no longer make economic profit that can go towards covering their entry cost. With faster generic entry rates, price drops to marginal cost much faster, meaning that there are far fewer quarters when generics can make economic profit, because there will be more competitors in the market right from the beginning. Thus, while it is true that generics can operate in the market longer due to being approved earlier, most of that time is going to be spent making near-zero economic profit.

Thus, for the fourth entrant, the prospects of entering the market have actually improved with the slower entry rate. Slowing down the entry rate creates an element of luck where an additional entrant can make large profits in the early periods of the market.

Next we rerun the counterfactuals but allow the AG to be released. Similar to before, for many market specifications, as generic entry rates increase, there are fewer generic entrants. However, with AG, the decline in generics as entry rate increases is smaller; that is, as entry rate increases, the number of generics declines by a smaller amount with AG presence compared to without AGs. In fact, there are more market-specifications for which there is no change in number of generics. The results are summarized in Table 11.

While faster entry rates lead to fewer generics, it has seemingly surprising effects on prices – prices are lower in the early periods of the market but are often higher in the later periods. Prices are lower in the early periods because, even though there are fewer generic entrants, most of them get released very quickly into the market after LOE. However, the fact that there are fewer generic entrants in total means that prices are often higher in the later periods of the market.

Table 7: Market outcomes with different generic entry rates for one molecule-formulation. The AG is prevented from entering the market for expositional clarity.

Cases	Total generics	Mean generic price (early periods)	Mean generic price (later periods)
Estimated	7.0	93.39	44.91
Estimated x 2	7.0	54.7	44.91
Immediate entry	6.0	53.28	53.28

Notes: Simulation conducted for market-specification of molecule-formulation Acamprosate-oral. “Estimated x n ” refers to the estimated generic entry rates multiplied by n . “Immediate entry” refers to all generics entering the market immediately after loss of exclusivity.

6.3 Ban on Authorized Generic

We examine if AGs (and similar fighting-brand strategies) should be encouraged or banned by policymakers. To do so, we simulate market specifications under allowing AG to be released versus AG ban. We find that a ban on AGs leads to weakly more generic entrants – and sometimes multiple additional new generics – but has mixed effects on overall prices.

We limit our counterfactuals to markets where over 90% of the simulations see an AG released. This is because it is only for these markets that the AG ban has an impact; in other market-specifications, if an AG is unlikely to enter, then imposing the ban makes little impact.

In our counterfactuals we find that, when AGs are banned, nearly all market specifications see an increase in the number of generics. In particular, over 60% of the markets see multiple additional new generics entering. For the large majority of market-specifications, AG ban leads to higher prices in the early periods of the market. If the AG was deterring multiple generics from entering, prices are lower in the later periods of the market with an AG ban. The exact effect on number of generics and prices depends on the specific demand and supply parameters. The results are summarized in Table 12.³¹

We first explain the impact from an AG ban on generic entrants. There are two opposing forces operating on different phases of the market's timeline after loss-of-exclusivity:

1. Early AG launch: Without an AG ban, the AG is released early in the market, so there is always an additional competitor in the market from the early quarters. Without the AG, the generics face lower competition in the initial periods as all the generics stochastically enter over time. Therefore, under an AG ban, prices are generally higher in the early periods of the market.
2. Greater generic entry: An AG ban leads to greater number of generic entrants. If there are two or more additional generic entrants, then prices are lower in the later periods of market.

The effect on later-period prices depends (among other things) on how many generics are deterred by the presence of an AG. If an AG ban leads to multiple new generics entering, then the resulting increase in competition more than offsets the missing price competition from the AG. As a result, in these markets the AG ban leads to lower prices in later periods. However if an AG ban leads to one or no new additional generic entrants, then AG ban leads to similar or higher prices in the later phase of the market .

Why can an AG ban lead to more than one new generic entering? This is because the AG has a larger negative impact on rivals' lifetime profits than an independent generic due

³¹We also simulate the impact of banning AG entry only for the first few periods (1-2 quarters). We find that the ban leads to smaller generic deterrence but qualitatively same impact on prices. The results are summarized in Table 13.

to its early-launch advantage. Since the AG is released early, it has a negative impact on every generic and particularly during the most profitable early periods of the market. If the AG is banned, not only is there one fewer competitor, but it is also one less competitor during the most profitable periods in the market. This results in a significant increase in expected generic profits compared to one fewer independent generic. In turn, this can incentivize multiple new generics to enter than if the AG had been present.

How many independent generics are deterred by the presence of the AG also depends on market-specific factors, particularly generic entry costs.

7 Conclusion

In the pharmaceutical industry, branded drug manufacturers can compete with generics by releasing an Authorized Generic, which is identical to the branded drug but without the brand label attached. Using quarterly drug sales and revenue data on US for 2004-2016, we estimate a structural model of pharmaceutical demand, generic entry, AG release, and pricing. We first set up and estimate a discrete choice model of demand where the consumer first chooses a group (brand, nonbrand, or outside option) then chooses a product within the group. Next we build a two-stage supply model of static generic entry followed by a dynamic AG release timing game. The model is solved using backward induction to recover equilibrium generic entrants and policy function of AG release. We estimate this model to recover supply-side parameters.

We simulate market equilibrium under different environments. First, we show how deterrence works in this setting - generics deter through being able to file for entry earlier (early-mover advantage) while AGs can deter because they can launch the product earlier. Next, we find that a faster entry rate leads to weakly fewer generic entrants. This is because greater competition upon launch offsets the advantage from an early launch. Finally, we show that a ban on AGs leads to weakly more generic entrants and mixed effect on overall prices. The latter happens because while there is increased price competition from weakly more generic entrants in the later stages of the market, it is counteracted by lack of an additional competitor through the AG in the early stages of the market. The exact effect depends on how many generics is being deterred by the AG; for a large number of market specifications, we find that the AG may deter multiple generics.

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A Model solution and cost estimation

A.1 Solving the full model

We solve the model by backward induction and recover the equilibrium number of generic entrants and policy function of authorized generics. We adapt the approach from [Igami \(2018\)](#) to our setting.

Note that the AG manufacturer knows the total generics approved in $t = 2$, i.e. $\mathcal{N}_2$, but can only form probabilistic beliefs about future generic launches. In contrast, generic manufacturers only know the equilibrium generic entrants n_e^* , not $\mathcal{N}_2$.

However, the CCP of the AG varies based on the exact value of $\mathcal{N}_2$, since they suggest different probability distributions over future generic launches. That is, the CCP of AG varies by $\mathcal{N}_2$.

The CCP of the AG is used to calculate the value function of the generics in stage 1. While the generics don't know the exact value of $\mathcal{N}_2$, they can form probabilistic beliefs about it using the FDA approval rates.

Thus, for the model solution, first we solve for the CCP of the AG and construct the corresponding generic value function for each $\mathcal{N}_2$. Then each generic value function (at the initial state s_1) is weighted by the probability of the corresponding $\mathcal{N}_2$ being realized. The resulting weighted generic value function is then used for the first stage.

The solution method is as follows:

1. For a given number of equilibrium generic entrants $\bar{n}$:
 - (a) For each $\mathcal{N}_2$ between 0 to $\bar{n}$:
 - i. Calculate the authorized generic's state transition matrices as determined by $\bar{n}$, T , FDA approval rates, and $\mathcal{N}_2$.
 - ii. Solve for policy function of authorized generic by backward induction from terminal period T . The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T .

- iii. Use the CCP, (10), T , and FDA approval rates to calculate the state transition matrices for generics.
 - iv. These state transition matrices are combined with per-period profits to construct value functions of generics.
 - v. Separately, calculate the probability of this $\mathcal{N}_2$ being realized using data on FDA approval rates.
- (b) For the initial state s_1 , find the generic value function by taking the probability-weighted average of the generic value functions for each $\mathcal{N}_2$ at the state s_1 . The probability of each $\mathcal{N}_2$ is being used as weights in this weighted average.
- (c) Denote this value function as $V(\bar{n})$ (other arguments suppressed).
2. Repeat the above steps for $\bar{n} = 1, \dots, N$ where N is a large integer.
 3. The equilibrium number of generics n^* is such that $V(n^*) \geq \kappa_m^g$ and $V(n^* + 1) < \kappa_m^g$. The value function is decreasing in $\bar{n}$.
 4. For this n^* re-solve the model to get the equilibrium policy function of authorized generics conditional on n^* and every possible $\mathcal{N}_2$.

Note that during simulations for a given market, we run many simulations for each possible $\mathcal{N}_2$, then report probability-weighted averages of each summary statistic (such as AG release fraction, AG release time, mean prices, etc.). The probability of each $\mathcal{N}_2$ is being used as weights in this weighted average. We rescale probabilities if some $\mathcal{N}_2$ see no AG released.

A.2 AG entry cost estimation

For AG cost estimation, we can skip a large chunk of the full solution method. From the data we observe n_{em}^* for each market m as well as $\mathcal{N}_{m2}$.

For a given value of parameters:

1. We solve for policy function of authorized generic by backward induction from terminal period T . The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T .

2. The choice probabilities are used to calculate the likelihood.

This process is repeated for different trial values of parameters until the objective function is minimized.

We estimate the model by Maximum Likelihood.

The contribution to the likelihood function from the observation of molecule-formulation m in period t is given by:

$$f(AG_{mt}|s_{mt}, \sigma_\varepsilon) = pr(AG_{mt} = 1)^{1(AG_{mt}=1)} pr(AG_{mt} = 0)^{1(AG_{mt}=0)}$$

In our estimation we set $T = 32$ quarters. However, many molecule-formulations are followed for much less than 32 quarters in our data. Thus, let $T_{max}(m)$ be the maximum number of quarters a molecule-formulation is followed.

The likelihood for observations of molecule-formulation m is given by

$$\mathcal{L}_m(\sigma_\varepsilon) = \prod_{t=0}^{T_{max}(m)} f(AG_{mt}|s_{mt}, \sigma_\varepsilon)$$

The joint likelihood over all molecule-formulations is given by

$$\mathcal{L}(\sigma_\varepsilon) = \prod_m \mathcal{L}_m(\sigma_\varepsilon)$$

During estimation, we maximize the log-likelihood to recover the parameter σ_ε .

$$\arg \max_{\sigma_\varepsilon} \log \mathcal{L}(\sigma_\varepsilon)$$

All other parameters informing the policy function in the likelihood function are either calculated offline or calibrated.

A.3 Estimation of marginal costs

Fix some trial value of generic entry cost κ_g across all molecule-formulations. We try $\kappa_g = \$3$ million and $\$15$ million, which are the publicly reported lower and upper bounds on generic's sunk entry costs. (Intuitively, the $\$3$ and $\$15$ million provide upper and lower

bounds on the marginal cost scaling parameter ϑ respectively, as we explain below.) For a given value of κ_g , we pick a grid of values for ϑ at 0.01 intervals between 0 and 1. For each value on the grid, solve for the equilibrium number of generic entrants.

The objective function is the sum of squared differences between the observed generic entrants in each market and the predicted entrants. One issue is that the model cannot predict less than zero entrants for a market (i.e. the outcome variable is censored at 0) and so the objective function may be minimized by predicting zero entrants for some markets. To tackle this problem, the objective function also includes a “censoring penalty”, where if a market is predicted to have zero generic entrants, it is assigned the censoring penalty value instead of the difference between observed and predicted entrants. This is to prevent the estimation from assigning zero entrants to lots of markets to minimize the objective function. In our estimation, we try censoring penalties of 3 and 4 and obtain similar results. We find the objective function to be convex over our grid of ϑ values.

Returning to the estimation, for a given κ_g , we calculate the objective function for every value on the grid. We pick the value that minimizes the objective function. This gives us the ϑ for that κ_g . We repeat this process for the different κ_g mentioned previously.

B Additional details on the demand model

B.1 Discussion on demand model

We now give a discussion of why we choose the two-stage demand setup. Papers estimating demand systems generally follow the discrete-choice setup in [Berry et al. \(1995\)](#), where each consumer chooses from a set of products. In contrast, in our setting, one agent chooses a group, then a different agent (such as the pharmacy) chooses a product within that group. This is because we found, after experimenting with different specifications, that this matched the empirical patterns of prices and market share best.

In our data we see that the price difference between the brand and non-brand group leads to substantial switching from branded drug to non-branded drugs, which implies high price elasticity of consumers. However, within non-brand products, we see very little switching based on specific prices of non-brand drugs. When all products are estimated together in a single discrete choice setting like [Berry et al. \(1995\)](#), we end up finding very low price-

elasticity overall for the market, which is at odds with the large brand-to-nonbrand substitution we see. This holds true even when we allow for random coefficients with different distributions and nested logit specifications.

From prior literature, the price difference between nonbrand drugs in the same market should be small. Since the price observed in our data is averaging over geographical variation, measurement errors, sales channels, and potential outliers, the price difference between nonbrands we observe in the data may be somewhat noisy.

Our specification gets around this by capturing the right substitution patterns in the data. Stage 1 captures the fact that the price gap between brand and nonbrand leads to a large substitution towards the cheaper nonbrand products. Stage 2 picks up the substitution between generics and AGs. While our demand model is somewhat different from other discrete-choice settings, note that our aim of the project is to study entry decisions; as long as we correctly predict prices and per-period profits, our central arguments work.³²

B.2 Demand estimation details

B.2.1 Estimation of First stage

In the First stage, the Consuming Agent chooses a group (brand, nonbrand, or outside option). We model this as a multinomial discrete-choice setting. Given our assumption of logit preference shocks, we can use the logit inversion from [Berry \(1994\)](#) to recover the demand parameters.

For $g = \text{brand or nonbrand}$, we estimate the following regression:

$$\ln(s_{gmt}) - \ln(s_{0mt}) = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} \ln p_g + \beta_1^{(1)} \text{brand}_g + \beta_2^{(1)} \text{brand}_g \cdot \text{time-since-loe} + \xi_g^{(1)} \quad (12)$$

B.2.2 Estimation of Second stage

In the second stage, the Distributing Agent takes as given the group chosen in the first stage and chooses a product within the group. If the group is brand or outside option, then the Distributing Agent has a single choice to make with probability 1. However if the group

³²Another option is to use a flexible regression on observed revenues to predict firm revenues in different states. An example of such a regression is presented in [Table 2](#).

chosen is nonbrand, then there are multiple products within the group, and so the Distributing Agent has to make a choice among them. We model this as another multinomial discrete-choice.

Given that idiosyncratic errors follow Type-1 Extreme Value distribution, we can write market share of generic drug j conditional on nonbrand group being chosen in Stage 1, as:

$$s_{jt|g} = \frac{e^{\delta_{jt}^{(2)}}}{\sum_k e^{\delta_{kt}^{(2)}}}$$

where $\delta_{kt}^{(2)} = \gamma_{m(k)}^{(2)} + \beta^{(2)}AG_k + \alpha^{(2)}\ln(p_{kt}) + \xi_{kt}^{(2)}$. Using the transformation from [Berry \(1994\)](#):

$$\ln(s_{jt|g}) = -\ln\left(\sum_k e^{\delta_{kt}^{(2)}}\right) + \delta_{jt}^{(2)}$$

Plugging in the formula for $\delta_{jt}^{(2)}$ gives us the estimating equation:

$$\ln(s_{jt|g}) = -\ln\left(\sum_k e^{\delta_{kt}^{(2)}}\right) + \gamma_{m(j)}^{(2)} + \beta^{(2)}AG_j + \alpha_2\ln(p_{jt}) + \xi_{jt}^{(2)} \quad (13)$$

That is, the first term on the right hand side varies at the market-by-quarter level, so including market-by-quarter fixed effects absorbs it. The remaining coefficients can just be estimated through a linear regression with fixed effects ($\xi_{jt}^{(2)}$ becomes the econometric error in the regression).³³

B.3 Results from BLP estimation

C Counterfactual results

For implementing the counterfactuals, we divide the 241 molecule-formulations into “small”, “medium”, and “large” markets. The “small” markets are assigned generic entry cost of \$1 million, “medium” markets are assigned per-period fixed cost of and generic entry cost of

³³By contrast, in [Berry \(1994\)](#) the authors use the outside good’s market share to net out the term $\ln(\sum_k e^{\delta_{kt}})$. In our case this cannot be done because there is no outside option in Stage 2; conditional on choosing the non-brand group in Stage 1, the consumer has to choose from one of the non-brand products in Stage 2.

\$3 million, and “large” markets are assigned per-period fixed cost of and generic entry cost of \$15 million .

Now we explain how we classify markets into these 3 groups. The general problem is that for the lowest entry cost of \$3 million, a number of molecule formulations predict 0 generic entrants or more than 22 generic entrants (the highest number observed in our dataset).

As a result, for each molecule-formulation, we try the three generic entry costs mentioned previously. A molecule-formulation is assigned to the group with the largest costs such that the model predicts positive generic entrants less than 22 and more than zero.

For 41 of the 241 molecule-formulations our model does not predict positive generic entrants, and we drop those markets from our counterfactuals. In the end, we have 31 molecule-formulations in the “small” markets, 163 molecule-formulations in the “medium” markets, and 6 molecule-formulations in the “large” markets.

Note that we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values.

A note on terminology: throughout this section “Early periods” refers to first 4 quarters, and “Later periods” refers to last 4 quarters (the terminal period is quarter 32).

D Discussion on AG’s choice-specific shocks

During estimation, these logit shocks are used to reconcile why we observe firms in similar states of the world make different choices. This is a common methodology used in dynamic discrete choice models with dollar-valued payoffs. Papers which have unitless payoff functions instead normalize the logit scaling parameter to 1.0 and estimate all other structural parameters relative to it, as in [Rust \(1987\)](#).

A more detailed list of such factors is as follows. First, this could be driven by manufacturing delays, organizational frictions, and contracting frictions (if AG production is being outsourced). It may be that the loss-of-exclusivity was unanticipated or there was uncertainty surrounding it, which meant that the branded manufacturer was unprepared for the loss-of-exclusivity when it happens. There could be policy uncertainty such as how the FDA would approve or regulate the new generic entrants, or how a lawsuit might be resolved, that may add further uncertainty to the release decision. It could be that the branded

manufacturer is engaging in other generic-detering strategies (such as product-line extensions) which affects its AG timing decision. The branded drug manufacturer may not have any generic subsidiaries, which makes it very difficult to release an AG, because marketing a non-branded drug versus a branded drug entail very different skillsets. The marketing of a branded drug focuses on advertising and detailing, negotiating formulary positions by insurer, etc. The marketing of a generic drug focuses on extreme cost efficiency, winning pharmacy auctions ([Cuddy \(2020\)](#)), etc. These are very different skillsets, and for a branded drug, it may not be worth the hassle of getting into the generics business just to release an authorized generic. It could instead outsource its production to another generic firm (for instance, Prasco and Greenstone specialize in bringing authorized generics for other firms into the market) but that involves negotiation cost and profit-sharing.

D.1 Early-move vs early-launch

Table 8: Distribution of outcomes across market-specifications.

Percentiles	AG release fraction
10%	0.0
20%	0.0
30%	0.0
40%	0.0
50%	0.0
60%	0.0
70%	0.0
80%	0.0
90%	0.0

Notes: For this counterfactual we fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. We find that AG almost never launches.

Table 9: Distribution of outcomes across market-specifications.

Percentiles	Difference in generics
10%	-2.0
20%	-1.0
30%	-1.0
40%	-1.0
50%	-1.0
60%	-1.0
70%	-1.0
80%	-1.0
90%	0.0

Notes: We vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG always enters in the period of LOE. Next we run simulations where the AG always enters 8 quarters after LOE. "Difference in generics" is the equilibrium generic entrants when the AG launches early minus the equilibrium generic entrants when the AG launches late. We find that across most market-specifications, the AG can deter more generics when it can launch earlier.

D.2 Faster generic approval rates

Table 10: Distribution of outcomes across market-specifications with estimated versus immediate generic approval. The AG is prevented from entering for expositional clarity.

Percentiles	Difference in no. of generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	0.0	0.115	-3.07
20%	0.0	0.256	-0.742
30%	0.0	0.387	-0.414
40%	1.0	0.619	-0.212
50%	1.0	0.891	-0.122
60%	1.0	1.463	-0.044
70%	1.0	3.337	0.0
80%	1.0	7.288	0.0
90%	2.0	23.612	0.0

Notes: “Difference in no. of generics” is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

Table 11: Distribution of outcomes across market-specifications with estimated versus immediate generic approval.

Percentiles	Difference in no. of generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	0.0	0.044	-1.486
20%	0.0	0.108	-0.216
30%	0.0	0.253	-0.086
40%	0.0	0.368	-0.008
50%	0.0	0.607	0.0
60%	0.0	0.98	0.0
70%	1.0	2.192	0.0
80%	1.0	6.113	0.0
90%	1.0	20.067	0.0

Notes: “Difference in no. of generics” is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

D.3 Ban on AG

Table 12: Distribution of outcomes across market-specifications.

Percentiles	Difference in no. of generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	-2.0	-0.54	-0.0
20%	-2.0	-0.271	0.0
30%	-2.0	-0.171	0.0
40%	-2.0	-0.088	0.018
50%	-2.0	-0.056	0.049
60%	-2.0	-0.025	0.118
70%	-1.0	-0.016	0.231
80%	-1.0	-0.003	0.411
90%	-1.0	0.112	1.066

Notes: “Difference in no. of generics” is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned. Other variables similarly defined. We find that banning AG generally leads to more generic entrants. We assume that in the absence of a ban, the AG would have drawn a trade shock and therefore could have chosen to enter.

Table 13: Distribution of outcomes across market-specifications.

Percentiles	Difference in no. of generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	-2.0	-4.691	0.0
20%	-1.0	-1.229	0.0
30%	-1.0	-0.494	0.024
40%	-1.0	-0.282	0.066
50%	-1.0	-0.203	0.121
60%	-1.0	-0.111	0.2
70%	-1.0	-0.068	0.309
80%	0.0	-0.04	0.675
90%	0.0	-0.02	1.892

Notes: “Difference in no. of generics” is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned for the first 2 quarters. Other variables similarly defined.

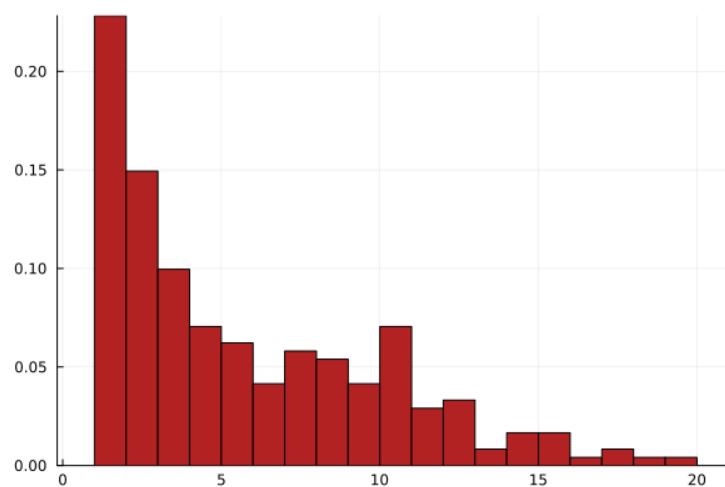
E Additional estimation results

F Additional summary statistics

Delay	Count
0.0	58
1.0	10
2.0	6
3.0	2
4.0	2
5.0	3
6.0	2
8.0	1
9.0	1
10.0	1
11.0	1
12.0	2
14.0	2
15.0	2
17.0	1
19.0	2
20.0	2
21.0	1
22.0	2
27.0	1
38.0	1
41.0	1

Table 14: AG Delay table. This shows, of the 110 AGs released, how many quarters after loss-of-exclusivity they were released. “Delay” is the number of quarters after loss-of-exclusivity. “Count” is the number of AGs released for the corresponding quarters since loss-of-exclusivity.

Figure 5: Histogram of total generic entry by molecule-formulation



Notes: This distribution has a 25th percentile of 2, median of 4, 75th percentile of 8, and 90th percentile of 11. There are 241 molecule-formulations.

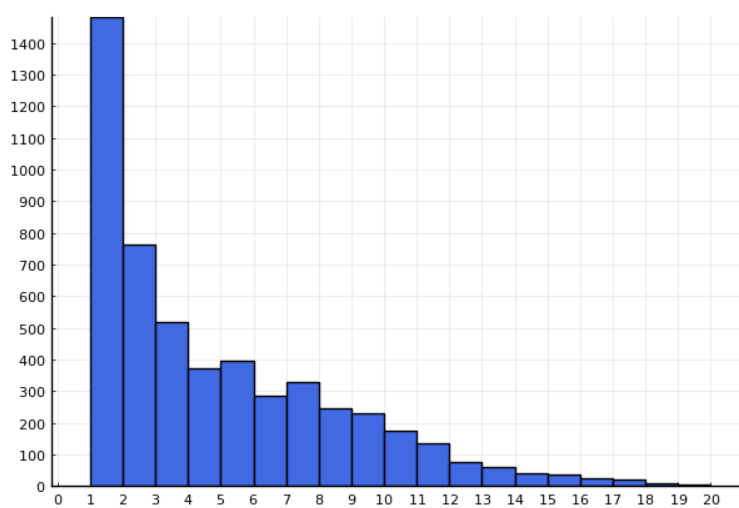


Figure 6: Histogram of count of generics present by molecule-formulation-quarter.

Table 15

	Percentiles	10%	25%	50%	75%	90%
Generic revenue		0.01	0.13	0.79	3.13	9.2
AG revenue		0.1	0.63	3.46	8.67	24.98
Brand revenue		0.18	0.87	4.41	22.08	99.1
Mean brand revenue		0.35	1.2	4.94	19.84	67.81
Mean generic revenue		0.29	0.82	2.35	5.36	12.33
Mean AG revenue		0.32	1.27	4.32	12.15	25.19
Mean brand within molform mktshare		3.51	6.21	15.61	34.92	69.76
Mean generic within molform mktshare		7.96	11.05	18.54	31.91	51.59
Mean AG within molform mktshare		4.19	11.57	22.04	36.01	49.7
Generic total lifetime revenue by product		0.26	1.73	11.06	54.78	184.7
AG total lifetime revenue by product		3.97	10.99	54.45	201.78	444.5

Notes:

G Miscellaneous

G.1 Links

1. The FDA list on AGs can be found here: <https://www.fda.gov/media/77725/download>.
2. The online source by AG producers can be found here: <https://www.authorizedgenerics.com/product-finder/>
3. The FDA list on Paragraph IV lawsuits can be found here: <https://www.fda.gov/media/133240/download>.

G.2 Comparison with **Fowler et al. (2023)**

One difference between **Fowler et al. (2023)** and our findings is they find AG incidence is just 6% of the markets, while we find nearly 44%. We believe this is mainly because they define product and possible markets for AG release (“at risk markets”) more broadly than us. In particular, differences in AG entry rate arise due to:

1. Definition of product: We consider a product market to be “molecule-formulation”, where formulation is either oral, injectable, or everything else. [Fowler et al. \(2023\)](#) define a product to vary by strength and more detailed dosage forms. As a result, their number of markets is much larger than ours.
2. At-risk markets: We consider a market to potentially see AG entry (at-risk market) if we find that it has seen at least one generic entry (and that first generic entry happened after 2004). Instead, [Fowler et al. \(2023\)](#) considers at-risk products to be “number of products approved during or after 1985 that had not been withdrawn or discontinued and that still had patent or exclusivity protection in each year from 2010 to 2019”. This is a very broad definition. Since AG release is only considered when a generic enters, our definition is consistent with our research question.
3. Undercounting AGs: [Fowler et al. \(2023\)](#) mentions that for data issues “... led to conservative estimates of the incidence of authorized generic entry”
4. Difference in yearly coverage: We look at 2004-2016, while they consider 2010-2019.

Other findings from [Fowler et al. \(2023\)](#) match our paper closely.

G.3 Comparison of generic approval rates with [Starc and Wollmann \(2023\)](#)

Another paper that measures FDA approval rates of generics is [Starc and Wollmann \(2023\)](#). Their setting and data is different from ours. [Starc and Wollmann \(2023\)](#) studies generic application and entry for markets where the loss-of-exclusivity has happened well in the past. They also have data on when the generic filed their ANDA, and so they can exactly measure the time between filing and approval. They report the distribution of "regulatory delays" in years and report it in a graph; numerically they find the following distribution:

- 1 year: 0.1
- 2 year: 0.34
- 3 year: 0.34

- 4 year: 0.15
- 5 year: 0.07
- 6 year: 0.04

(These figures are read off a graph and so aren't exact.)

Compared to our estimates, they find much slower ANDA approval rates. We are not sure why that is. One reason could be that we are modeling markets which face loss-of-exclusivity. For some of these markets, the loss-of-exclusivity is anticipated, and so generics may start filing in advance. As a result, they arrive on the market much sooner compared to the setting in [Starc and Wollmann \(2023\)](#), where generics notice a cartel formation/high prices due to cartel formation and then apply for ANDA approval.

As a result we think that while [Starc and Wollmann \(2023\)](#) calculates approval rates that are correct for their setting, they might be too slow for our setting.

One weakness to our approach is that we do not have data on when generics file their ANDAs, and so we cannot directly measure the time between filing and approval. Instead what we do is calculate the rate at which generics launch in the market and use it to recover the approval rate.

An alternative is we can re-estimate our model and counterfactuals using the ANDA approval rates found in [Starc and Wollmann \(2023\)](#).

H Media coverage of AGs

1. <https://www.forbes.com/sites/ikebrannon/2024/01/25/how-drug-companies-use-authorized-generics-to-keep-drug-prices-high/>
2. <https://kffhealthnews.org/news/drugmakers-now-masters-at-rolling-out-their-own-generics-to-stifle-competition/>
3. <https://www.americanprogress.org/article/removing-obstacles-to-generic-drug-competition/>
4. <https://www.precisionvaluehealth.com/2019/04/drug-markers-get-good-pr-via-authorized-generics-but-other-effects-are-less-clear/>

5. <https://www.biosimilardevelopment.com/doc/will-authorized-biologics-disrupt-the-market-for-biosimilars-0001>

I Useful quotes from FTC (2011)

I.1 Quotes on how the brand manufacturer tightly controls the timing of AGs

- (Pg 73)“Early launch, however, “cannibalizes” sales of the brand-name product, starting their decline before ANDA-generic entry has occurred. Brand-name firms have chosen to tightly control the timing of AG launch to avoid unnecessary loss of brand sales and other disadvantages. Launch even a short time before confirmation of ANDA-generic entry is disfavored, and AG supply agreements often include strict penalty provisions or financial terms that enforce launch timing.”
- (Pg 73 footnote) “See CD, Nov. 4, 2004 (“financial risk to brand [of early launch is] significant” because brand erosion cannot be “made up by benefit of 2 week early launch”; that risk is exacerbated if the first generic entrant is delayed for any reason, because the AG would cannibalize the brand during the delay); CD, Feb. 2, 2005 (“Do not put Rx brand at risk (i.e., do not launch in advance of a generic, but can have contracts in place and inventory at wholesalers”)) (emphasis original); CD, Jan. 5, 2007 (“[generic subsidiary] and [brand-name company] agree it is not financially viable to launch an AG prior to launch of other AB rated generics in the US”); CD, Mar. 31, 2004 (disadvantages of launching the AG early include “Uncertainty over ultimate LOE date [that] could cause unnecessary brand erosion; Additional incentive for earlier ‘at risk’ launches by generics; Potential public relations issue; Legislative Risks”).”
- “(Pg 21) The agreements also establish the timing of AG launch. Many agreements provide for launch only after ANDA-generic entry or the date of expected ANDA-generic launch. To ensure that brand sales are not eroded before ANDA-generic competition begins, penalty provisions often apply if the AG marketer launches before generic entry”

I.2 Quotes on outsourcing AG production

- (Pg 77)“Profit-splits, i.e., the percentage of profits that the AG distributor must pay to the brand, are also consistent with the brand’s interest in the revenue stream arising from an AG. Apart from contexts where AG marketing rights derive from patent litigation settlements, marketing agreements generally require AG distributors to pay the brand a large percentage of profits on the AG, typically from 50-92 percent.”
- (Pg 85) “At the same time, some documents also suggest that AG distribution opportunities for generic firms even as early as 2004 were becoming more difficult to obtain, thereby prompting some to re-visit their AG strategies.³⁰ As one generic firm document explains: AGx opportunities and expected returns have both been substantially reduced in recent months, due to the decision of various brand companies to market more of their products through their own AGx subsidiaries, and increased competition among generic companies for available AGx deals...

Revenue and earnings shortfalls by many publicly-traded generic companies due to severe pricing competition has made AGs one of the only ways to maintain growth in the short term; The resulting intense competitive bidding for remaining opportunities has made brand companies realize they are in a position to retain most of the economic value of authorized generic versions of their intellectual property; Historically, generic partners received 20-35% of the economics, but this has now been driven down to 10% or less; The reduced value of AG deals for generic companies will hurt the business models of some companies...”

- “(Pg 23-24) A number of terms affect potential profitability: • Pricing: Neither settlement nor non-settlement agreements restrict the generic company’s authority in setting the price of AGs. Indeed, they frequently affirm that authority. • Transfer Price: The transfer or supply price establishes the amount the generic company pays the brand-name company for the AG. In some agreements, this price is related by express terms to the cost of manufacturing the AG.³⁵ • Profit-Splits: Most agreements provide a profit-split, which governs the distribution of profits between the brand and generic companies from sales of the AG. Many non-settlement agreements require the AG marketer to pay the brand a large percentage of profits on the AG.³⁶

Profit-splits have implications for generic firms' incentives: although AG distribution agreements can give a generic company that was not the first filer the benefit of being able to market during the 180-day exclusivity period, first-filing status provides the possibility of exclusivity unburdened by a profit-split, albeit with the likely expense of patent litigation."

I.3 Quotes on how AGs are usually not aimed at 180-day

- "(Pg 27) Although drug companies marketed many AGs during 180-day exclusivity, they marketed more during non-exclusive periods, and often for drugs that had not been subject to a patent challenge. "
- "(Pg 32) Taken together, the exclusivity and patent certification data in Figures 2-8 and 2-10 indicate that during the period when AGs became common, undermining 180-day exclusivity was unlikely to have been the primary rationale for many AG drugs. This does not preclude a disincentive effect or rationale for those AGs that were marketed during 180-day exclusivity periods, but it suggests that different or additional explanations for AG marketing also should be considered. "
- "(Pg 32) Nonetheless, AGs marketed during exclusivity were in the minority. More AGs were marketed after exclusivity or when no exclusivity had occurred; many of the latter drugs had not been subject to a patent challenge. The rate of AG launch increased as pre-entry brand sales increased, and within most sales levels was similar for ANDA-generic entry with and without exclusivity; there was no consistent pattern in the relative magnitude of rates of AG entry for generic entry with and without exclusivity. These findings suggest that, on average, the anticipated sales level of the brand-name drug has had greater bearing on decisions to launch AGs than whether ANDA-generic entry occurred via exclusivity. The findings, however, do not preclude a disincentive rationale or effect for AGs that were marketed during exclusivity."